Novell Pharmaceutical Laboratories

Effective & Safe Fever & Pain Management in Children: Ibuprofen Syrup & Paracetamol Infusion Solutions

Barbi E, Marzuillo P, Neri E, Naviglio S, Krauss BS. Fever in Children: Pearls and Pitfalls. Children (Basel). 2017;4(9):81. Published 2017 Sep 1. doi:10.3390/children4090081
Sullivan JE, Farrar HC. Fever and Antipyretic Use in Children. Pediatrics. 2011;127(3). doi:10.1542/peds.2010-3852

Burden of Fever and Pain in Pediatric Practice

Demam adalah salah satu keluhan anak tersering di IGD / Klinik → juga merupakan salah satu pendorong utama kunjungan & kecemasan keluarga ortu

Nyeri sering disertai infeksi akut, inflamasi lokal (tonsillitis/otitis), pasca operasi, trauma minor

Dampak klinis:
Distress anak (discomfort, sleep disruption, intake berkurang)
Beban caregiver (work-loss, repeated visits)
Risiko overuse/misuse antipiretik (dosis double, interval pemberian terlalu deket)

Merlo F, Falvo I, Caiata-Zufferey M, et al. New insights into fever phobia: a pilot qualitative study with caregivers and their healthcare providers. Eur J Pediatr. 2023;182(2):651-659. doi:10.1007/s00431-022-04704-4
Clericetti CM, Milani GP, Bianchetti MG, et al. Systematic review finds that fever phobia is a worldwide issue among caregivers and healthcare providers. Acta Paediatrica. 2019;108(8):1393-1397. doi:10.1111/apa.14739
Trippella G, Ciarcià M, de Martino M, Chiappini E. Prescribing Controversies: An Updated Review and Meta-Analysis on Combined/Alternating Use of Ibuprofen and Paracetamol in Febrile Children. Frontiers in Pediatrics. 2019;7. doi:10.3389/fped.2019.00217

Fever Phobia

Fobia demam adalah kekhawatiran berlebihan tentang konsekuensi demam pada anak-anak

Efek Fever Fobia:

- Overdosing / terlalu cepat redosing (interval dipersingkat, kejar angka suhu turun cepat ke normal)
- Terapi kombinasi tanpa indikasi jelas
- Kunjungan/rujuk/rawat inap yang sebenarnya tidak perlu (karena panik suhu, bukan kondisi klinis)

Table 1 Factors possibly associated with parental fever phobia

	Significance
Low educational and socioeconomic status	$p < 0.0001$
Ethnicity (Latin Americans, Bedouins, Turkish, Southern Italians)	$p < 0.001$
Past history of febrile seizures	$p < 0.0001$
Young maternal age	$p < 0.0001$

Barbi E, Marzuillo P, Neri E, Naviglio S, Krauss BS. Fever in Children: Pearls and Pitfalls. Children (Basel). 2017;4(9):81. Published 2017 Sep 1. doi:10.3390/children4090081

Definition & Classification of Fever in Children

Fever: kenaikan suhu tubuh akibat set point hipotalamus meningkat (ini termasuk respons terkontrol)

Hyperthermia: suhu naik karena kegagalan termoregulasi (set point tidak naik) → potensi lebih berbahaya

Target terapi simptomatik:
Kurangi discomfort, bukan turunkan angka suhu ke normal
Antipiretik → anak tampak distress/nyeri/poor intake

Table 1 NICE guidelines for identifying low-risk, intermediate-risk and high-risk fever in children [2]

	Green/low risk	Amber/intermediate risk	Red/high risk
Color (of skin, lips or tongue)	Normal color	Pallor reported by parent/carer	Pale/mottled/ashen/blue
Activity	Responds normally to social cues Content/smiles Stays awake or awakens quickly Strong normal cry/not crying	Not responding normally to social cues No smile Wakes only with prolonged stimulation Decreased activity	No response to social cues Appears ill to a healthcare professional Does not wake or if roused does not stay awake Weak, high-pitched or continuous cry
Respiratory		Nasal flaring Tachypnea: respiratory rate >50 breaths/minute, age 6–12 months >40 breaths/minute, age >12 months Oxygen saturation ≤95% in air Crackles in the chest	Grunting Tachypnea: respiratory rate >60 breaths/minute Moderate or severe chest indrawing
Circulation and hydration	Normal skin and eyes Moist mucous membranes	Tachycardia: >160 beats/minute, age <12 months >150 beats/minute, age 12–24 months >140 beats/minute, age 2–5 years Capillary refill time ≥3 seconds Dry mucous membranes Poor feeding in infants Reduced urine output	Reduced skin turgor
Other	None of the amber or red symptoms or signs	Age 3–6 months and temperature ≥39 °C Fever for ≥5 days Rigors Swelling of a limb or joint Non-weight bearing limb/not using an extremity	Age <3 months and temperature ≥38 °C Non-blanching rash Bulging fontanelle Neck stiffness Status epilepticus Focal neurological signs Focal seizures

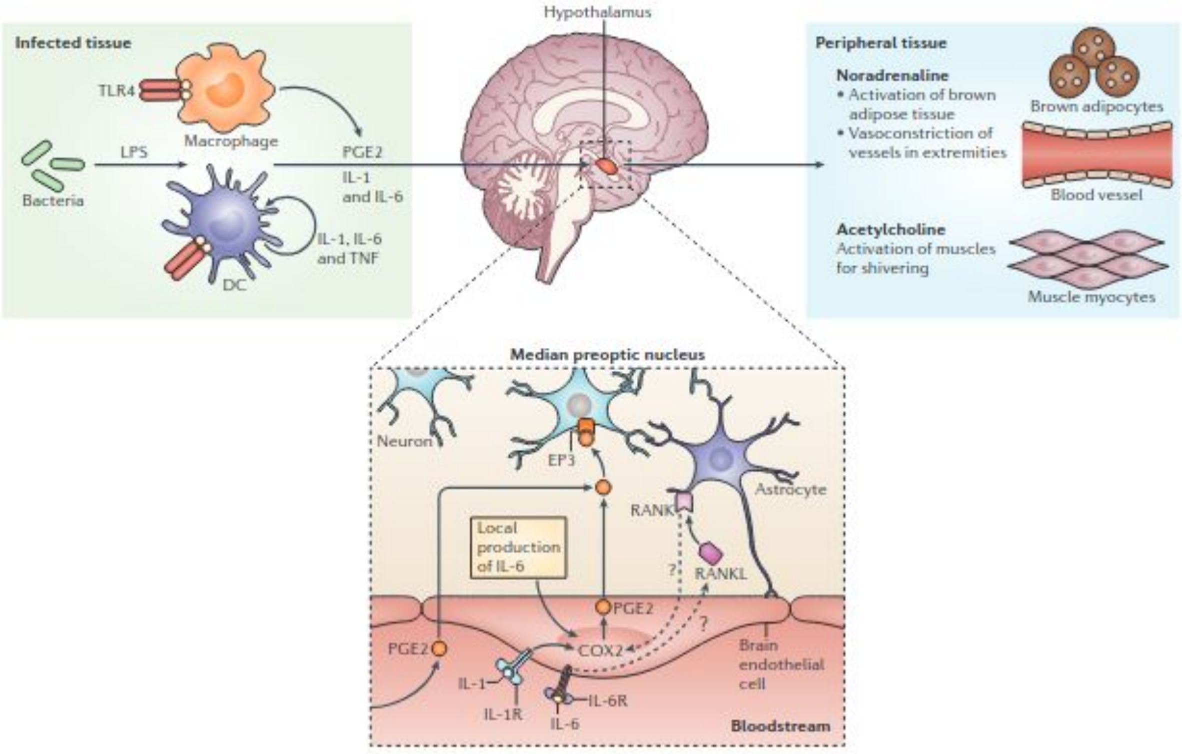
NICE National Institute for Health and Care Excellence

Kanabar D. A practical approach to the treatment of low-risk childhood fever. Drugs R D. 2014;14(2):45-55. doi:10.1007/s40268-014-0052-x

Classification of Fever in Children

Evans SS, Repasky EA, Fisher DT. Fever and the thermal regulation of immunity: the immune system feels the heat. Nature Reviews Immunology. 2015;15(6):335-349. doi:10.1038/nri3843

Pathophysiology of Fever



Infeksi mengaktifkan TLR4 pada makrofag/DC → pelepasan sitokin pirogen (IL-1β, IL-6, TNF) + pembentukan PGE2 di jaringan perifer

Sitokin beredar ke otak → endotel otak mengekspresikan COX-2 → meningkatkan produksi PGE2

PGE2 berikatan dengan reseptor EP3 di area preoptik/hipotalamus → set-point suhu naik (termostat tubuh diset lebih tinggi)

Hipotalamus mengaktifkan respons efektor:

- Noradrenalin (simpatis) → vasokonstriksi + aktivasi brown adipose tissue (termogenesis)
- Asetilkolin (motorik) → menggigil (shivering)

Produksi & konservasi panas meningkat → suhu tubuh naik (demam) hingga pirogen/PGE2 turun dan set-point kembali normal

Sansone L, Gentile C, Grasso EA, et al. Pain Evaluation and Treatment in Children: A Practical Approach. Children (Basel). 2023;10(7):1212. Published 2023 Jul 13. doi:10.3390/children10071212
Eccleston C, Fisher E, Howard RF, et al. Delivering transformative action in paediatric pain: a Lancet Child & Adolescent Health Commission. Lancet Child Adolesc Health. 2021;5(1):47-87. doi:10.1016/S2352-4642(20)30277-7
de Sévaux JLH, et al. Paracetamol (acetaminophen) or non-steroidal anti-inflammatory drugs, alone or combined, for pain relief in acute otitis media in children. Cochrane Database Syst Rev. 2023;8:CD011534. doi:10.1002/14651858.CD011534.pub3

Pediatric Pain

Nyeri adalah salah satu keluhan tersering pada anak di klinik & layanan akut (sering muncul pada berbagai diagnosis: muskuloskeletal, headache, abdominal pain, otalgia, sore throat)

Nyeri sering underrecognized & undertreated → efeknya: distress, takut prosedur, compliance buruk

Efek inadequate acute pain control jangka panjang → perubahan struktur/konektivitas otak terkait paparan nyeri akut periode perinatal dan berkorelasi dengan efek kognitif–perilaku di usia dewasa

Eccleston C, Fisher E, Howard RF, et al. Delivering transformative action in paediatric pain: a Lancet Child & Adolescent Health Commission. Lancet Child Adolesc Health. 2021;5(1):47-87. doi:10.1016/S2352-4642(20)30277-7

Types of Pediatric Pain

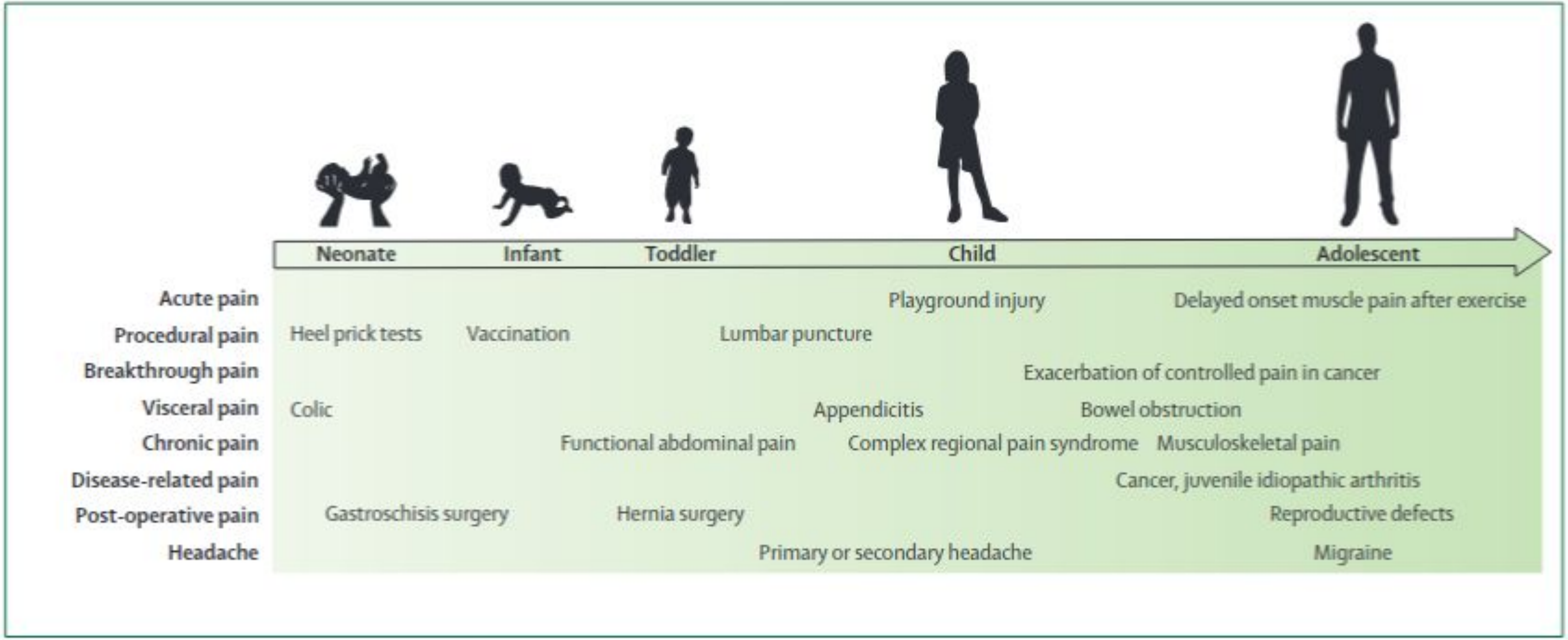


Figure 1: **Common types of pain during childhood**

Sansone L, Gentile C, Grasso EA, et al. Pain Evaluation and Treatment in Children: A Practical Approach. Children (Basel). 2023;10(7):1212. Published 2023 Jul 13. doi:10.3390/children10071212

Types of Pain Mechanism

	Nociceptive Pain		Neuropathic Pain
	Somatic Pain	Visceral Pain	
Origin	Superficial receptors	Visceral receptors	Nerves
Location and distribution	• Superficial (skin and subcutaneous tissue) • Well-localized • Not referred	• Deep (e.g., from muscle/bone/fascia/periosteum) • Poorly localized • Often referred	• Nerve structures (e.g., trigeminal nerve) • Radiating or specific
Transmission	A-delta-fibers	C-fibers	Dermatomal (periphery) or non-dermatomal (central)
Type of pain reported	Pinprick, sharp or stabbing	Pressure, sharp or ache	Tingling, prickling, lancinating or burning

Pediatric Pain Assessment

Category	Scoring		
	0	1	2
Face	No particular expression or smile, eye contact and interest in surroundings	Occasional grimace of rown, withdrawn, disinterested, worried look to face, eyebrows lowered, eyes partially closed, cheeks raised, mouth pursed	Frequent to constant frown, clenched jaw, quivering chin, deep furrows on forehead, eyes closed, mouth opened, deep lines around nose / lips
Legs	Normal position or relaxed	Uneasy, restless, tense, increased tone, rigidity, intermittent flexion / extension of limbs	Kicking or legs drawn up, hypertonicity, exaggerated flexion / extension of limbs, tremors
Activity	Lying quietly, normal position, moves easily and freely	Squirming, shifting back and forth, tense, hesitant to move, guarding, pressure on body part	Arched, rigid, or jerking, fixed position, rocking, side to side head movement, rubbing of body part
Cry	No cry or moan (awake or asleep)	Moans or whimpers, occasional cries, sighs, occasional complaint	Crying steadily, screams, sobs, moans, grunts, frequent complaints
Consolability	Calm, content, relaxed, does not require consoling	Reassured by occasional touching, hugging or talking to, distractible	Difficult to console or comfort

Figure 1. The Face, Legs, Activity, Cry and Consolability (FLAAC) scale for pediatric pain.

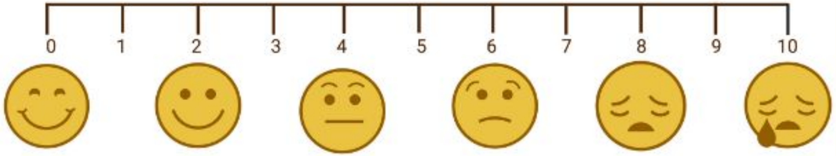


Figure 2. The Wong–Baker FACES scale for pediatric pain.

Table 2. The most often used pain scales in pediatrics.

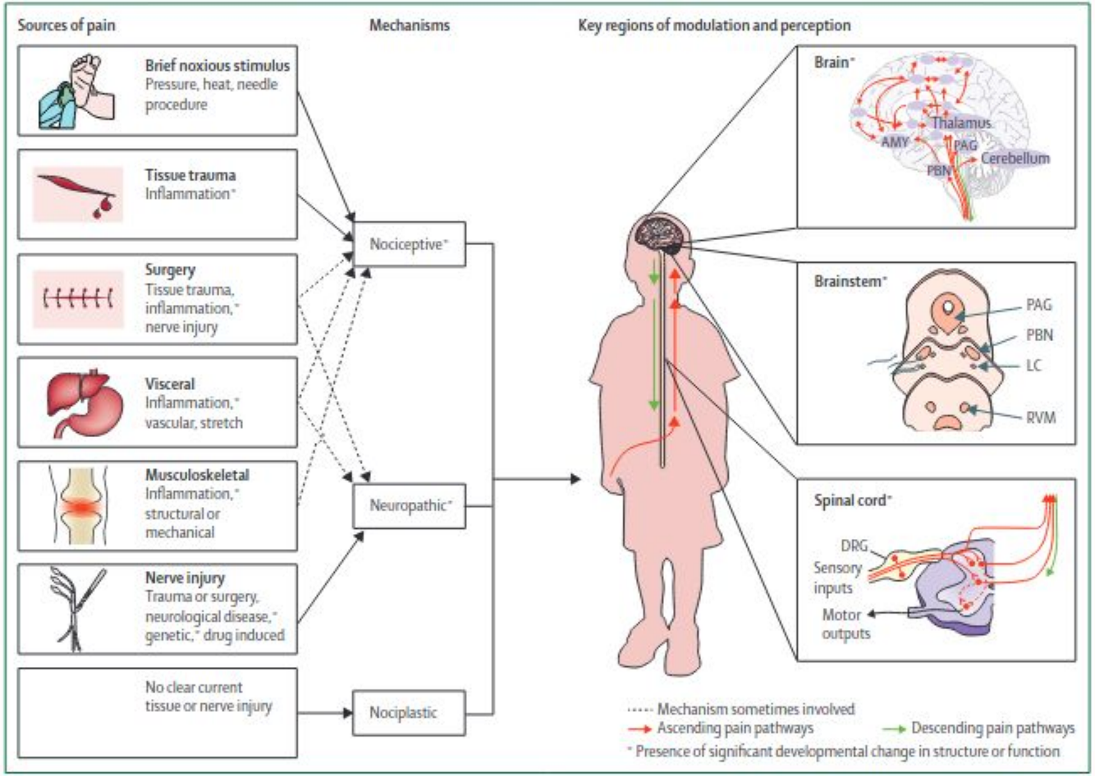
Acronym	Age Range
CHIPPS	Under 5 years
CHEOPS	1–7 years
FLACC	2 months–7 years
OPS and MOPS	8 months–13 years
Poker Chip Tool	From 3 years
Oucher Scale	3–12 years
Wong–Baker FACES® Pain Rating Scale	From 3 years
FPS-R	From 4 years
VAS	From 5 years
NRS	From 8 years

Abbreviations: CHIPPS (Children and Infants Postoperative Pain Scale); CHEOPS (Children's Hospital of Eastern Ontario Pain Scale); FLACC (Face, Legs, Activity, Cry and Consolability); OPS/MOPS (Objective Pain Scale/Modified Objective Pain Scale); FPS-R (Faces Pain Scale—Revised); VAS (Visual Analogue Scale); and NRS (Numeric Rating Scale).

Sansone L, Gentile C, Grasso EA, et al. Pain Evaluation and Treatment in Children: A Practical Approach. Children (Basel). 2023;10(7):1212. Published 2023 Jul 13. doi:10.3390/children10071212

Eccleston C, Fisher E, Howard RF, et al. Delivering transformative action in paediatric pain: a Lancet Child & Adolescent Health Commission. Lancet Child Adolesc Health. 2021;5(1):47-87. doi:10.1016/S2352-4642(20)30277-7
Sansone L, Gentile C, Grasso EA, Di Ludovico A, La Bella S, Chiarelli F, Breda L. Pain Evaluation and Treatment in Children: A Practical Approach. Children. 2023; 10(7):1212. https://doi.org/10.3390/children10071212

Pathophysiology Pain in Children



Sumber nyeri: jaringan/inflamasi (nociceptive), saraf (neuropathic), atau perubahan pemrosesan nyeri tanpa bukti kerusakan jelas (nociplastic).

Jalur ascending: nociceptor → saraf perifer → kornu dorsalis → thalamus → korteks (persepsi).

Modulasi: sistem limbik (cemas/distress) + jalur descending dapat memperkuat/melemahkan nyeri.

Pada anak, distress/ketakutan sering memperberat persepsi nyeri

Section on Clinical Pharmacology and Therapeutics; Committee on Drugs, Sullivan JE, Farrar HC. Fever and antipyretic use in children. Pediatrics. 2011;127(3):580-587. doi:10.1542/peds.2010-3852
Whyte LA, Al-Araji RA, McLoughlin LM. Guidelines for the management of acute gastroenteritis in children in Europe Archives of Disease in Childhood - Education and Practice 2015;100:308-312.

Disease-Based Approach

Demam & nyeri = gejala → selalu pikirkan penyakit dasar + red flags

pilih antipiretik/analgesik berdasar

- Distress
- Rute pemberian
- Komponen inflamasi
- Kontraindikasi

Kondisi umum demam / nyeri pada anak:

- OMA
- Tonsilofaringitis
- Influenza/ILI
- Gastroenteritis akut

de Sévaux JLH, Damoiseaux RA, van de Pol AC, et al. Paracetamol (acetaminophen) or non-steroidal anti-inflammatory drugs, alone or combined, for pain relief in acute otitis media in children. Cochrane Database Syst Rev. 2023;8(8):CD011534. Published 2023 Aug 18. doi:10.1002/14651858.CD011534.pub3
Lieberthal AS, et al. Clinical Practice Guideline The Diagnosis and Management of Acute Otitis Media.Pediatrics. 2013;131(3):e964–e999

Otitis Media Akut

OMA: otalgia + demam + iritabel.

AAP (AOM severe): otalgia sedang–berat / otalgia ≥48 jam atau suhu ≥39ºC.
Terapi awal: kontrol nyeri/demam dulu, lalu keputusan antibiotik sesuai kriteria & usia

TABLE 3 Treatments for Otalgia in AOM

Treatment Modality	Comments
Acetaminophen, ibuprofen[63]	Effective analgesia for mild to moderate pain. Readily available. Mainstay of pain management for AOM.
Home remedies (no controlled studies that directly address effectiveness)	May have limited effectiveness.
Distraction	
External application of heat or cold	
Oil drops in external auditory canal	
Topical agents	
Benzocaine, procaine, lidocaine[65,67,70]	Additional, but brief, benefit over acetaminophen in patients older than 5 y.
Naturopathic agents[68]	Comparable to amethocaine/phenazone drops in patients older than 6 y.
Homeopathic agents[71,72]	No controlled studies that directly address pain.
Narcotic analgesia with codeine or analogs	Effective for moderate or severe pain. Requires prescription; risk of respiratory depression, altered mental status, gastrointestinal tract upset, and constipation.
Tympanostomy/myringotomy[73]	Requires skill and entails potential risk.

American Academy of Pediatrics
DEDICATED TO THE HEALTH OF ALL CHILDREN™

Organizational Principles to Guide and Define the Child Health Care System and/or Improve the Health of all Children

CLINICAL PRACTICE GUIDELINE

The Diagnosis and Management of Acute Otitis Media

Pelucchi C, Grigoryan L, Galeone C, et al. Guideline for the management of acute sore throat. Clinical Microbiology and Infection. 2012;18:1-27. doi:10.1111/j.1469-0691.2012.03766.x

Tonsilofaringitis

Tonsilofaringitis (viral / bakterial) → demam + nyeri tenggorok/odinofagia
Target terapi awal → perbaikan gejala + cari penyebab

ESCMID PUBLICATIONS 10.1111/j.1469-0691.2012.03766.x

Guideline for the management of acute sore throat

ESCMID Sore Throat Guideline Group

Either ibuprofen or paracetamol are recommended for relief of acute sore throat symptoms (A-I).

de Sévaux JLH, Damoiseaux RA, van de Pol AC, et al. Paracetamol (acetaminophen) or non-steroidal anti-inflammatory drugs, alone or combined, for pain relief in acute otitis media in children. Cochrane Database Syst Rev. 2023;8(8):CD011534. Published 2023 Aug 18. doi:10.1002/14651858.CD011534.pub3
Friedman MJ, Attia MW. Clinical Predictors of Influenza in Children. Archives of Pediatrics & Adolescent Medicine. 2004;158(4):391. doi:10.1001/archpedi.158.4.391
Ebell MH, Afonso A. A systematic review of clinical decision rules for the diagnosis of influenza. Ann Fam Med. 2011;9(1):69-77. doi:10.1370/afm.1192
Section on Clinical Pharmacology and Therapeutics; Committee on Drugs, Sullivan JE, Farrar HC. Fever and antipyretic use in children. Pediatrics. 2011;127(3):580-587. doi:10.1542/peds.2010-3852

ARTICLE

Clinical Predictors of Influenza in Children

Marla J. Friedman, DO; Magdy W. Attia, MD

Table 2. Univariate Analysis of Clinical Features of Influenza-Positive Patients

Finding	Frequency of Reported Symptom, %	OR (95% CI)
Historical		
Fever, ≥39°C	55	2.4 (1.2-5.0)
Rhinorrhea	60	0.8 (0.4-1.7)
Cough	83	2.2 (1.0-5.2)
Vomiting	36	1.6 (0.7-3.4)
Diarrhea	10	0.8 (0.3-2.2)
Decreased intake	57	1.0 (0.5-2.0)
Decreased activity	54	1.8 (0.8-3.6)
Headache*	44	2.6 (1.2-5.8)
Abdominal pain and/or nausea*	31	3.2 (1.2-8.2)
Sore throat	37	1.5 (0.7-3.2)
Myalgia*	33	0.8 (0.4-1.8)
Apnea†	0	0.5 (0.0-3.9)
Clinical		
Coryza	57	1.6 (0.8-3.2)
Cough (observed)	33	1.3 (0.6-2.9)
Wheezing	5	0.3 (0.1-1.0)
Rales	3	0.5 (0.1-2.4)
Retractions	0	0.1 (0.0-0.6)
Nasal flaring	0	0.2 (0.0-1.2)
Rash	10	1.6 (0.3-8.1)
Conjunctivitis	3	0.6 (0.1-3.7)
Pharyngitis	35	2.0 (0.9-4.3)
Cervical adenopathy	14	1.7 (0.6-5.4)
Otitis media	10	1.0 (0.3-3.3)
Abdominal tenderness	12	1.8 (0.5-6.0)

Influenza / Influenza-like illness

Influenza/ILI: demam, batuk, sakit tenggorok, sakit kepala, fatigue, myalgia.

Clinical predictor saat outbreak: triad batuk + sakit kepala + faringitis membantu prediksi influenza (sensitivitas 80%, spesifisitas 78%)

Antipiretik diberikan untuk comfort/distress, bukan target normalisasi suhu

Pelucchi C, Grigoryan L, Galeone C, et al. Guideline for the management of acute sore throat. Clinical Microbiology and Infection. 2012;18:1-27. doi:10.1111/j.1469-0691.2012.03766.x
Section on Clinical Pharmacology and Therapeutics; Committee on Drugs, Sullivan JE, Farrar HC. Fever and antipyretic use in children. Pediatrics. 2011;127(3):580-587. doi:10.1542/peds.2010-3852

GEA (Gastroenteritis Akut)

GEA: diare dan/atau muntah ± demam → terapi utama: rehidrasi & koreksi dehidrasi

Antipiretik/analgesik = suportif untuk comfort

muntah/dehidrasi/risiko AKI → utamakan paracetamol
Hindari ibuprofen sampai hidrasi aman

Fields, E., Chard, J., Murphy, M. S., & Richardson, M. (2013). Assessment and initial management of feverish illness in children younger than 5 years: summary of updated NICE guidance. BMJ, 346(may22 16), f2866–f2866. https://doi.org/10.1136/bmj.f2866
Cettler M, Zielińska M, Rosada-Kurasińska J, Kubica-Cielińska A, Jarosz K, Bartkowska-Śniatkowska A. Guidelines for treatment of acute pain in children - the consensus statement of the Section of Paediatric Anaesthesiology and Intensive Therapy of the Polish Society of Anaesthesiology and Intensive Therapy. Anaesthesiol Intensive Ther. 2022;54(3):197-218. doi:10.5114/ait.2022.118972
Kanabar D. A practical approach to the treatment of low-risk childhood fever. Drugs R D. 2014;14(2):45-55. doi:10.1007/s40268-014-0052-x

Antipyretic & Analgesic Options in Pediatrics

Untuk demam anak, 2 antipiretik yang paling digunakan:
Parasetamol (acetaminophen) & ibuprofen
Tujuan utama adalah mengurangi distress

Untuk nyeri akut → non-opioid (parasetamol/NSAID) sebagai fondasi multimodal
Opioid → nyeri sedang–berat/indikasi khusus

Kombinasi/alternating parasetamol–ibuprofen → bisa meningkatkan kontrol gejala dibeberapa studi
Tetapi kombinasi/alternating menambah kompleksitas & risiko error dosis

Fatan FA, Hilmi IL, Salman. An Overview of Selection of Antipyretic Drugs for Children. Journal of Pharmaceutical and Science. 2023;6(1):230-236.
Fields, E., Chard, J., Murphy, M. S., & Richardson, M. (2013). Assessment and initial management of feverish illness in children younger than 5 years: summary of updated NICE guidance. BMJ, 346(may22 16), f2866–f2866. https://doi.org/10.1136/bmj.f2866
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Vittinghoff M, Lönnqvist PA, Mossetti V, et al. Postoperative pain management in children: Guidance from the pain committee of the European Society for Paediatric Anaesthesiology (ESPA Pain Management Ladder Initiative). Paediatr Anaesth. 2018;28(6):493-506. doi:10.1111/pan.13373

Guidelines for Paracetamol Options in Pediatrics

AAP merekomendasikan parasetamol untuk demam anak, fokus untuk comfort
Juga menekankan risiko overdosis

Guideline NICE: Parasetamol sebagai opsi untuk anak demam bila distress

Guideline nyeri akut pediatrik menempatkan acetaminophen (oral/rectal/IV) sebagai bagian multimodal analgesia

Guideline perioperatif (ESPA Pain Management Ladder) → acetaminophen rute IV sebagai komponen multimodal pasca operasi pada anak

Fatan FA, Hilmi IL, Salman. An Overview of Selection of Antipyretic Drugs for Children. Journal of Pharmaceutical and Science. 2023;6(1):230-236.
Fields, E., Chard, J., Murphy, M. S., & Richardson, M. (2013). Assessment and initial management of feverish illness in children younger than 5 years: summary of updated NICE guidance. BMJ, 346(may22 16), f2866–f2866. https://doi.org/10.1136/bmj.f2866
Mitchell RB, Archer SM, Ishman SL, et al. Clinical Practice Guideline: Tonsillectomy in Children (Update)-Executive Summary. Otolaryngol Head Neck Surg. 2019;160(2):187-205. doi:10.1177/0194599818807917

Guidelines for Ibuprofen Options in Pediatrics

AAP: ibuprofen sebagai antipiretik/analgesik efektif pada anak

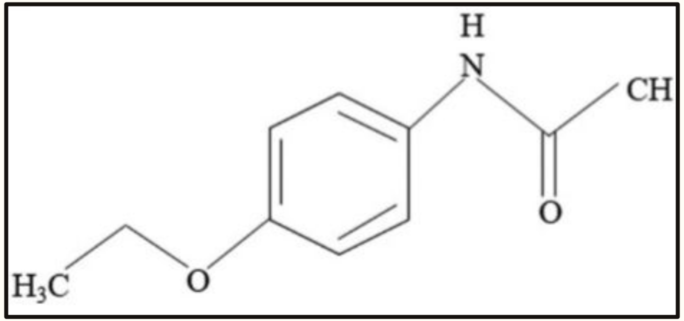
Guideline NICE: ibuprofen sebagai opsi untuk anak demam bila distress

Drug interventions to reduce body temperature

1.6.4 Consider using either paracetamol or ibuprofen in children with fever who appear distressed. **[2013]**

Guideline tonsilektomi anak merekomendasikan ibuprofen (dan/atau acetaminophen) untuk kontrol nyeri pasca tindakan

Fatan FA, Hilmi IL, Salman. An Overview of Selection of Antipyretic Drugs for Children. Journal of Pharmaceutical and Science. 2023;6(1):230-236.



Paracetamol

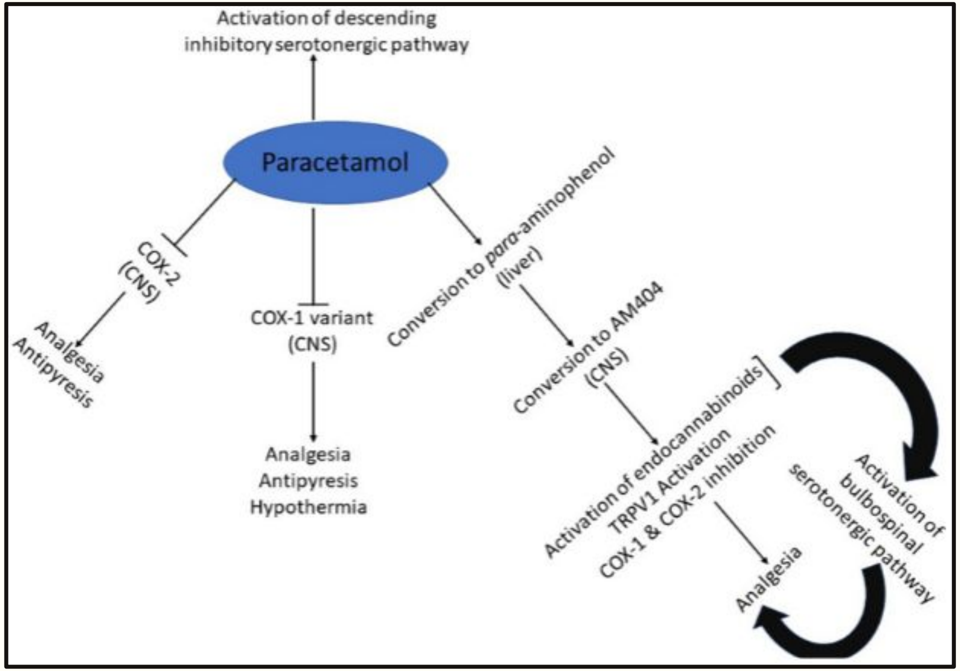
Parasetamol (acetaminophen) adalah analgesik–antipiretik non-opioid yang sangat luas digunakan pada anak → untuk demam dan nyeri ringan–sedang

AAP merekomendasikan dosis berbasis berat badan dan menekankan pencegahan error dosis

Dosis PO/PR/IV: 10–15 mg/kg/dosis tiap 4–6 jam sesuai kebutuhan

Ayoub SS. Paracetamol (acetaminophen): A familiar drug with an unexplained mechanism of action. Temperature. 2021;8(4):351-371. doi:10.1080/23328940.2021.1886392

Mechanism of Action Paracetamol



Efek utama sentral (CNS): menghambat jalur COX di otak → menurunkan prostaglandin → antipiretik + analgesik

Aktivasi jalur desenden serotonergik: paracetamol membantu mengaktifkan descending inhibitory serotonergic pathway → menekan transmisi nyeri di spinal cord

paracetamol di hati jadi p-aminophenol → di CNS dikonversi menjadi AM404 → aktivasi endocannabinoid & TRPV1 → kontribusi efek analgesik

Paracetamol Single Dose vs Multidose

Farmakokinetik IV acetaminophen bersifat dose-proportional → 500 mg/50 mL lebih sesuai untuk kebutuhan dosis anak yang berbasis berat badan

Right-size vial for right-size patient

Patient weight	Dose per administration	Volume per administration	Maximum volume of Perfalgan (10 mg/mL) per administration based on upper weight limits of group (mL)***	Maximum Daily Dose ***
≤10 kg *	7.5 mg/kg	0.75 mL/kg	7.5mL	30 mg/kg
>10 kg to ≤33kg	15 mg/kg	1.5mL/kg	49.5mL	60mg/kg not exceeding 2g
>33 kg to ≤50kg	15 mg/kg	1.5mL/kg	75 mL	60mg/kg not exceeding 3g

50 mL vial → term newborn, infant, toddler, dan anak <33 kg
100 mL vial dibatasi untuk pasien >33 kg
>10 kg – ≤33 kg, dosis per pemberian 15 mg/kg = 1.5 mL/kg → volume maksimum 49.5 mL (pas dengan 50ml vial)

Tabor A, Shalemariam Z, Alemu Y, Gorems K. Bacterial contamination of single and multiple-dose parenteral injection vials after opening and antibiotic susceptibility of isolates at Jimma Medical Center, Jimma, Southwest Ethiopia. Infect Prev Pract. 2023;5(3):100290. Published 2023 Jun 10. doi:10.1016/j.infpip.2023.100290
Vincent F, Thompson J, Gray L, Bradberry S, Sandilands E, Thanacoody R, Tuthill D. Medication errors involving intravenous paracetamol in children: experience from enquiries to the National Poisons Information Service. Arch Dis Child. 2024 Jun 19;109(7):582-585. doi: 10.1136/archdischild-2023-326460. PMID: 38233098.
Veyckemans F, Anderson BJ, Wolf AR, Allegaert K. Intravenous paracetamol dosage in the neonate and small infant. British Journal of Anaesthesia. 2014;112(2):380-381. doi:https://doi.org/10.1093/bja/aet559

Problem with Multiple Dose

Studi cross-sectional menunjukkan single-dose maupun multiple-dose vial dapat mengalami kontaminasi bakteri setelah digunakan sekali

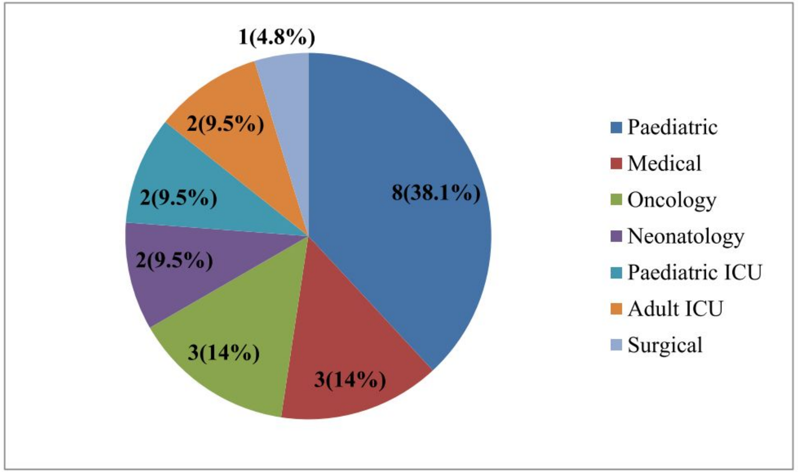


Figure 1. Distributions of contamination among ward and the intensive care unit at JMC, Jimma, Southwest Ethiopia, from July 2021 to October 2021.

Analisis NPIS UK dari 266 kasus error lebih sering pada anak kecil, 16,9% berupa 10-fold overdose dan 24,1% berupa pemberian oral + IV bersamaan tanpa sengaja

Analisis NPIS UK dari 266 kasus error lebih sering pada anak kecil, 16,9% berupa 10-fold overdose dan 24,1% berupa pemberian oral + IV bersamaan tanpa sengaja

Error of administration

I.V. paracetamol formulation strength is 10 mg ml^{-1}. Confusion between mg and ml is a well-known cause of drug administration error. The reports of i.v. paracetamol overdose are

Bucak I, Almis H, Dogan C, Turgut M. The status of drug wastage in the pediatric emergency department of a tertiary hospital. Avicenna Journal of Medicine. 2020;10(1):10. doi:https://doi.org/10.4103/ajm.ajm_58_19

Paracetamol 500 mg/50 mL Lebih Efisien

Studi di pediatric emergency department menunjukkan bahwa paracetamol 1000 mg vial memiliki mean wastage ratio 0.79, dengan rerata obat terpakai hanya 207.14 mg dan sisa 792.86 mg per vial

Table 1: Data obtained in the study

Drug name	Drug quantity (mg)- form	Mean quantity of drug used (mg)	Mean quantity of drug unused (mg)	Mean quantity of drug unused /drug quantity form	Cost of unused drugs (US$)	Total cost (US$)
Ondansetron	4-mg ampoule	1.56	2.44	0.61	57.75	94.52
Methylprednisolone	20-mg ampoule	9.9	10.1	0.5	121.11	242.22
	40-mg ampoule	30	10	0.25	1.01	4.05
Pheniramine maleate	45.5-mg ampoule	22.5	23	0.5	7.12	14.24
Dexamethasone	8-mg ampoule	4.67	3.33	0.42	21.31	50.77
Ampicillin-sulbactam	1000-mg vial	689.62	310.38	0.31	13.84	44.67
Midazolam	5-mg vial	1.84	3.16	0.63	10.43	16.56
	15-mg vial	8	7	0.47	0.29	0.62
Ranitidine	50-mg ampoule	34.26	15.74	0.31	1.48	4.78
Metamizole	1000-mg ampoule	462.77	537.23	0.54	2.43	4.51
Ceftriaxone	1000-mg vial	692.85	307.15	0.3	6.31	21.04
Diclofenac	75-mg ampoule	40	35	0.47	0.97	2.08
Paracetamol	1000-mg vial	207.14	792.86	0.79	17.99	22.78
Levetiracetam	500-mg vial	262.85	237.15	0.47	4.97	10.58
Cefazolin	1000-mg vial	575	425	0.42	3.83	9.13
Phenytoin	250-mg ampoule	174	76	0.3	3.43	11.45
Metoclopramide	10-mg ampoule	4	6	0.6	0.52	0.87
Vitamin K1	2-mg ampoule	1	1	0.5	2.07	4.14
Dimenhydrinate	50-mg ampoule	13.3	36.7	0.73	0.9	1.24
Omeprazole	40-mg vial	17	23	0.57	7.13	12.51
Cefuroxime	750-mg vial	600	150	0.2	0.33	1.67
Valproate Sodium	400-mg ampoule	200	200	0.5	2.95	5.91
Diazepam	10-mg ampoule	8	2	0.2	0.12	0.64
					280.09	580.98

Formulaso 500 mg lebih preferable untuk pasien pediatrik → karena menurunkan waste & kerugian biaya

HIGHLIGHTS of PRESCRIBING INFORMATION. https://www.accessdata.fda.gov/drugsatfda_docs/label/2018/022450s011lbl.pdf
Perfalgan 10 Mg/Ml, Solution for Infusion Paracetamol. Accessed April 13, 2026. https://assets.hpra.ie/products/Human/12798/2202377.PA0002_069_001.b6302ffc-91c4-4965-bbac-53793ff30c9c.000001PIL.180216.pdf
https://pdf.hres.ca/dpd_pm/00069718.PDF

Paracetamol Small Dose Lebih Cocok untuk Single Use

Paracetamol IV → single-dose container
unused portion must be discarded
Dan setelah seal ditembus dosis harus diberikan dalam 6 jam

volume of the commercially available container. The entire 100 mL container of acetominophen is not intended for use in patients weighing less than 50 kg. Acetominophen is supplied in a single-dose container and the unused portion must be discarded.

For single use only. The product should be used immediately after opening. Any unused solution should be discarded.

ACETAMINOPHEN INJECTION is for single-use only. Discard the unused portion.

Vial 500 mg/50 mL lebih menguntungkan → lebih kecil jadi kemungkinan menyisakan juga lebih kecil
Kalau sisa dari volume besar yang secara aturan justru harus dibuang

CDC. Safe Injection Practices to Prevent Transmission of Infections to Patients. Injection Safety. Published May 20, 2024. https://www.cdc.gov/injection-safety/hcp/clinical-guidance/index.html

Benefit of Single Dose Vial

Lebih baik untuk patient safety
CDC menyatakan single-dose vial lebih dipilih bila memungkinkan

clinic. The primary breaches in infection control practice that contributed to these outbreaks were 1) reinsertion of used needles into a multiple-dose vial or solution container (e.g., saline bag) and 2) use of a single needle/syringe to administer intravenous medication to multiple patients. In one of these

Penggunaan lebih sederhana secara operasional: tidak ada repeated access berkali kali, tidak ada dating opened vialnya

Okereke B, Ibeleme O, Bisi-Onyemaechi A. Randomised comparative trial of the efficacy of paracetamol syrup and dispersible tablets for the treatment of fever in children. J Int Med Res. 2021;49(3):300060521999755. doi:10.1177/0300060521999755

Paracetamol pada Demam anak

Desain: RCT komparatif paracetamol sirup vs paracetamol dispersible

Populasi: anak 6 bulan–12 tahun,suhu timpani >37,5ºC

Intervensi: Paracetamol 15 mg/kg sebagai single oral dose

Hasil: idak ada perbedaan signifikan penurunan suhu pada kedua kelompok di 30 menit & 60 menit
Penurunan suhu serupa di kedua kelompok
Tidak ada efek samping/ADR yang teramati atau dilaporkan selama periode observasi 1 jam

Table 4. Comparison of the number of febrile participants in each arm of the study.

	Paracetamol formulation			
	Dispersible, n (%)	Syrup, n (%)	χ^2	P value
T1				
Fever (≥37.5°C)	27 (100.0)	25 (100.0)	NA	NA
No fever	0 (0.0)	0 (0.0)		
T2				
Fever (≥37.5°C)	18 (66.7)	18 (72.0)	0.173	0.677
No fever	9 (33.3)	7 (28.0)		
T3				
Fever (≥37.5°C)	25 (92.6)	23 (92.0)	0.006	0.936
No fever	2 (7.4)	2 (8.0)		

T1, baseline temperature; T2, 30 minutes; T3, 60 minutes.
NA = Not Applicable.

Roy S, Simalti AK. Comparison of Antipyretic Efficacy of Intravenous (IV) Acetaminophen versus Oral (PO) Acetaminophen in the Management of Fever in Children. Indian J Pediatr. 2018;85(1):1-4. doi:10.1007/s12098-017-2457-3

Paracetamol IV vs Oral pada Demam anak

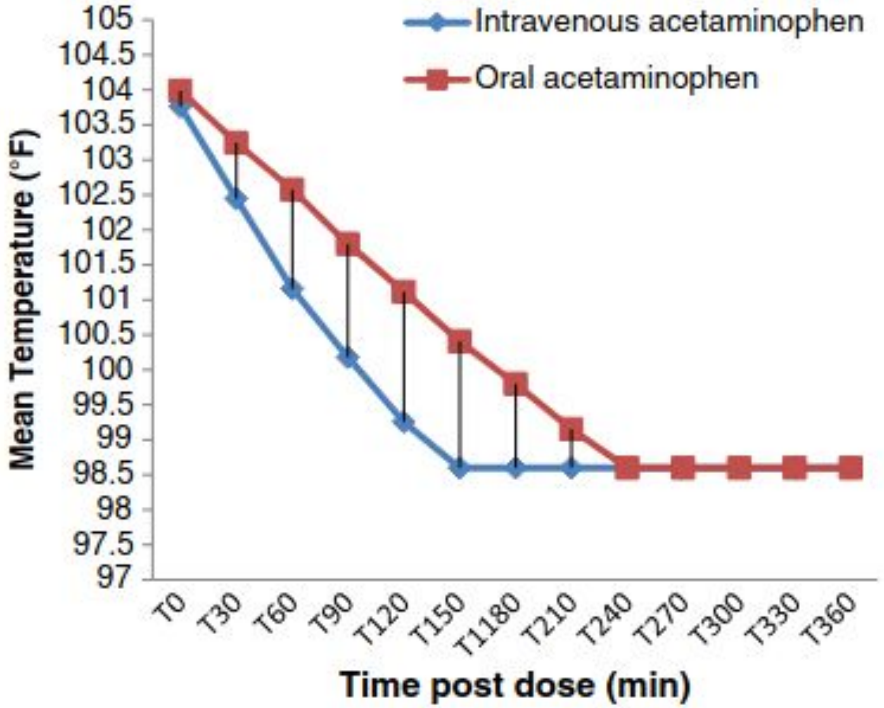


Fig. 1 Time post dose mean temperature variation in the study groups

Desain: Randomized controlled trial

Populasi: 400 anak demam; 200 kelompok IV 200 orak

Intervensi: IV acetaminophen 15 mg/kg/dosis sekali vs oral acetaminophen 15 mg/kg/dosis sekali, evaluasi selama 6 jam

Hasil: Penurunan suhu lebih cepat signifikan sampai 180 menit pada kelompok IV
Kebutuhan dosis tambahan juga lebih rendah pada IV (3% vs 5%)

Gunawan PI, Saharso D. Efficacy and Tolerability of Intravenous Paracetamol Compared to Oral Paracetamol for the Treatment of Childhood Fever. Pharmacognosy Journal. 2022;14(5):537-541. doi:https://doi.org/10.5530/pj.2022.14.132

Paracetamol IV vs Oral pada Demam anak

Table 3: The mean temperature reduction from baseline.

	Temperature reduction iv paracetamol	Temperature reduction oral paracetamol	p
t0 - t1	-0.142 (±0.339)	-0.023 (±0.228)	0.038*
t0 - t2	-0.644 (±0.420)	-0.262 (±0.372)	0.000*
t0 - t3	-0.908 (±0.476)	-0.462 (±0.453)	0.000*
t0 - t4	-1.090 (±0.579)	-0.660 (±0.600)	0.000*
t0 - t5	-1.446 (±0.678)	-1.094 (±0.816)	0.019*
t0 - t6	-1.394 (±0.750)	-1.192 (±0.842)	0.204
t0 - t7	-1.180 (±0.784)	-1.053 (±0.862)	0.439

*Significant at p <0.05 by independent sample t test

Desain: Randomized controlled, open-label clinical trial

Populasi: 104 anak demam usia >6 bulan

Intervensi: IV paracetamol 10 mg/kgBB vs oral paracetamol 10 mg/kgBB

Hasil: Suhu IV lebih rendah dan penurunannya lebih besar; perbedaan signifkan ada di 30 menit (p=0,005), 45 menit (p=0,002), dan 60 menit (p=0,006)
Efek samping sebanding

Zeilmaker-Roest G, de Vries-Rink C, van Rosmalen J, et al. Intermittent intravenous paracetamol versus continuous morphine in infants undergoing cardiothoracic surgery: a multi-center randomized controlled trial. Crit Care. 2024;28(1):143. Published 2024 Apr 30. doi:10.1186/s13054-024-04905-3

Paracetamol vs Morphine pada Pasca Operasi

Table 4 Primary and secondary study outcomes

Outcome	Paracetamol (n = 94)	Morphine (n = 100)	P-value
Cumulative morphine consumption, (Median (IQR)), mcg/kg*	145.0 (115.0–432.5)	692.6 (532.7- 856.1)	< 0.001
Rescue morphine dose total, median (IQR), mcg/kg*	29.4 (0–45.7)	30.0 (0–70.9)	0.38
Patients with rescue morphine bolus, (n) (%)*	62 (66.0%)	69 (69.0%)	0.76
Patients with rescue morphine infusions (n)*	40(42.3%)	42 (42.0%)	1.00
Co-medication (n)*			
Midazolam	76 (80.9%)	77 (77%)	0.59
Lorazepam	5 (5.3%)	10 (10%)	0.29
Propofol	26 (27.7%)	19 (19%)	0.18
Fentanyl	8 (8.5%)	10 (10%)	0.81
Clonidine	6 (6.4%)	5 (5%)	0.67
Ketamine	23 (24.5%)	21 (21.0%)	0.54
Dexmedetomidine	6 (6.4%)	8 (8%)	0.68
PICU stay in days (median(IQR))	3 (2–6)	2 (1–5)	0.10
Duration postoperative mechanical ventilation (median, IQR, hours)	11.8 (4.5–45.6)	15.4 (5.7–28.8)	0.39
Reintubation (n, %)*	3 (3.2)	2 (2)	0.68
Adverse events**			
Hemodynamic instability (bradycardia or hypotension)	22 (23.4%)	28 (28%)	0.46
Gastrointestinal (obstruction, obstipation, vomiting)	3 (3.2%)	1 (1%)	0.28
Delirium	3 (3.2%)	5 (5%)	0.53
Apnea	3 (3.2%)	2 (5%)	0.60
VIS	5 (0–9.6)	5 (3.0–10)	0.21
NRS ≥ 4 at least once (n,%)*	55 (59%)	62 (62%)	0.62
Comfort-B scale scores (median, range)*	12 (8–22)	12 (7–20)	0.05

*48 h postoperative

** 96 h postoperative

PICU = Pediatric Intensive Care Unit, NRS = Numeric Rating Scale, VIS = Vasoactive Inotropic Score

Desain: Multi-center, randomized, double-blind, controlled trial

Populasi: 194 anak usia 0–3 tahun pasca operasi jantung dengan cardiopulmonary bypass

Intervensi: IV paracetamol intermiten sebagai analgesik utama vs morphine kontinu

Hasil: Konsumsi morphine kumulatif 48 jam di grup IV paracetamol 79% lebih rendah signifikann dibanding grup morphine (145,0 vs 692,6 mcg/kg; P < 0,001)

Osorio D, Maldonado D, Rijs K, van der Marel C, Klimek M, Calvache JA. Efficacy of different routes of acetaminophen administration for postoperative pain in children: a systematic review and network meta-analysis. Efficacité des différentes voies d'administration de l'acétaminophène pour la douleur postopératoire chez les enfants : une revue systématique et méta-analyse en réseau. Can J Anaesth. 2024;71(8):1103-1116. doi:10.1007/s12630-024-02760-y

Paracetamol pada Nyeri Anak

Desain: systematic review + network meta-analysis membandingkan rute acetaminophen pada nyeri pascaoperasi pediatrik

Populasi: 14 RCT, 829 anak usia 30 hari–17 tahun

Intervensi: Paracetamol 15 mg/kg sebagai single oral dose

Hasil:
Perbandingan efektivitas nyeri: Oral lebih baik vs rektal (−0,88; −1,44 s/d −0,31), sedangkan IV vs oral tidak berbeda signifikan (−0,60; −1,20 s/d 0,01) dan IV vs rektal tidak berbeda signifikan (−0,28; −0,62 s/d 0,06)
Penggunaan oral / rectal / IV berdasarkan kondisi anak

Table 2 League table of pairwise comparisons in network meta-analysis for primary outcome pain assessment at 0–2 hours

0.99*		
Oral route	0.49*	
Difference in means, −0.60 (95% CI, −1.20 to 0.01)	**Intravenous route**	0.03*
Difference in means, −0.88 (95% CI, −1.44 to −0.31)	Difference in means, −0.28 (95% CI, −0.62 to 0.06)	**Rectal route**

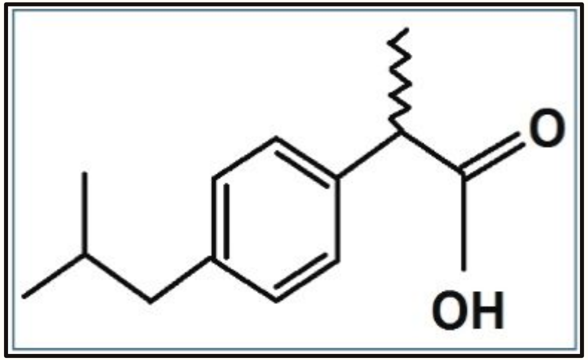
*P-Scores representing the ranking of treatments in frequentist network meta-analysis. They measure the mean extent of certainty that a treatment is better than the competing treatments.[23]

Average treatment estimates are provided as mean differences with 95% CIs*

*Quantifying heterogeneity / inconsistency: $tau^2 = 0.0685$; $tau = 0.2618$; $I^2 = 60.7\%$; 95% CI, 0.0%; 85.3%

CI = confidence interval

Rainsford KD. Ibuprofen: pharmacology, efficacy and safety. Inflammopharmacology. 2009;17(6):275-342. doi:10.1007/s10787-009-0016-x
Bushra R, Aslam N. An Overview of Clinical Pharmacology of Ibuprofen. Oman Medical Journal. 2010;25(3):155-161. doi:10.5001/omj.2010.49

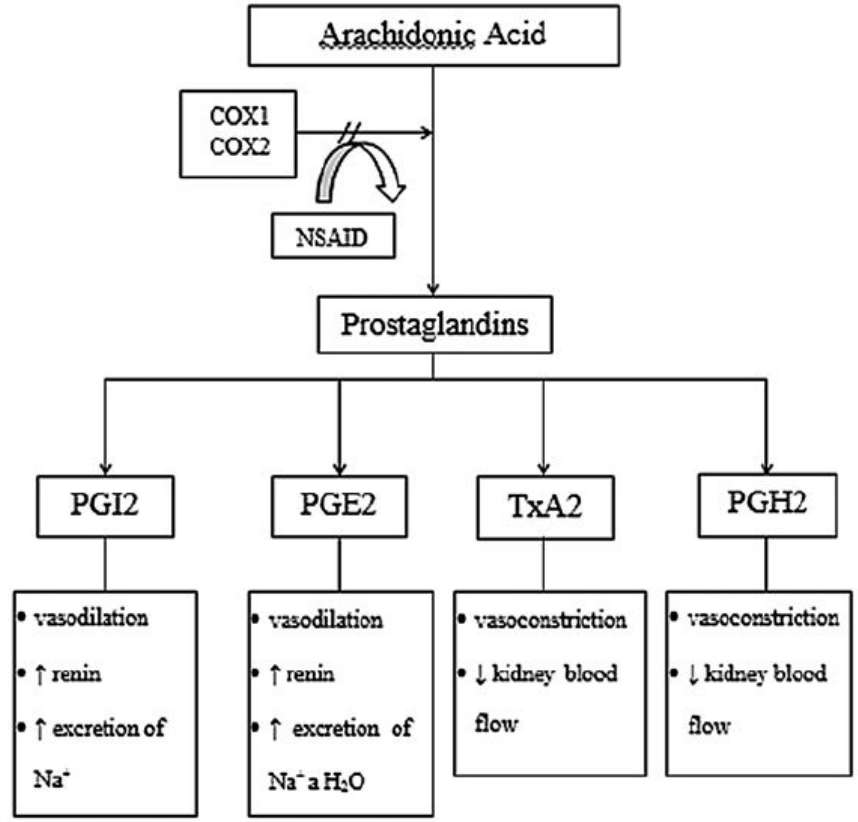


Ibuprofen

Ibuprofen adalah NSAID dengan efek analgesik, antipiretik, antiinflamasi

Dosis PO syrup: 5–10 mg/kg/dosis tiap 6–8 jam sesuai kebutuhan

Gargya I, Singh B, Talnia S. NSAIDS (Non- Steroidal Anti- Inflammatory Drugs)- Their Effects and Side Effects in Orthodontic Therapy- A Review. Dental Journal of Advance Studies. 2017;05(01):008-013. doi:10.1055/s-0038-1672075



COX 1, COX 2 = cyclooxygenase 1, 2; NSAID = non-steroidal anti-inflammatory drugs; PGI2 = prostacyclin; PGE2 = prostaglandin E2; TxA2 = thromboxane A2; PGH2 =

Mechanism of Action Ibuprofen

Ibuprofen = NSAID non-selektif → menghambat COX-1 & COX-2

INhibisi COX menurunkan konversi arachidonic acid → prostaglandins (PGE2/PGI2) dan TxA2

↓ PGE2 → antipiretik + analgesik

↓ prostaglandin perifer → anti-inflamasi (turun sensitisasi nociceptor)

Orlando BJ, Lucido MJ, Malkowski MG. The structure of ibuprofen bound to cyclooxygenase-2. J Struct Biol. 2015;189(1):62-66. doi:10.1016/j.jsb.2014.11.005

Additional Effect of Ibuprofen as Anti-Inflammatory

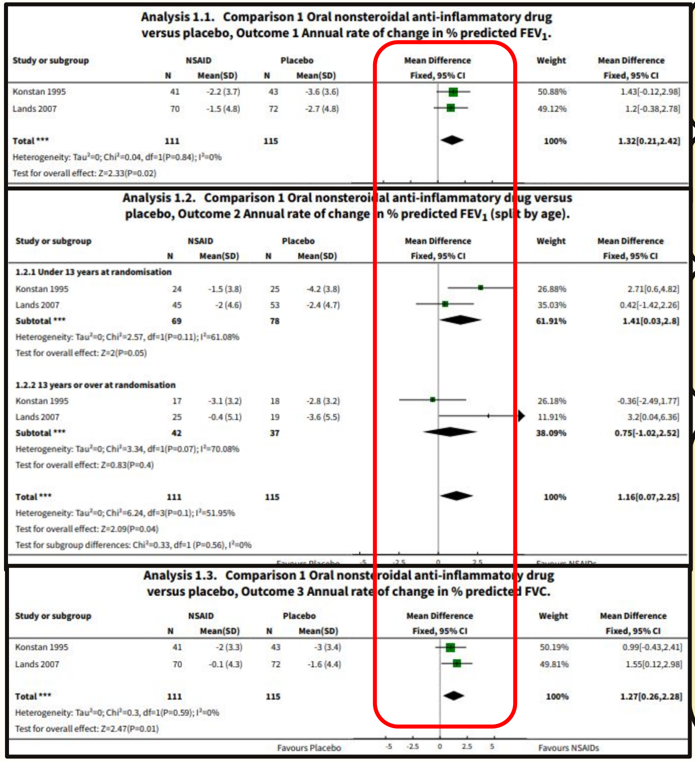
Reversible, non-selective COX inhibition (COX-1 & COX-2) → ↓ konversi asam arakidonat menjadi prostaglandins (terutama PGE_2)

Efek ↓ PGE_2 di jaringan inflamasi:
↓ vasodilatasi & permeabilitas vaskular → ↓ edema
↓ sensitisasi nociceptor perifer → ↓ inflammatory pain

↓ PGE_2 di sistem saraf pusat → ↓ amplifikasi sinyal nyeri (central sensitization) → analgesia lebih kuat pada nyeri inflamasi

Utsumi S, Amagasa S, Moriwaki T, Uematsu S. Oral analgesic for musculoskeletal injuries in children: A systematic review and network meta-analysis. Acad Emerg Med. 2024; 31: 61-70. doi:10.1111/acem.14803

Ibuprofen sebagai Anti-Inflamasi pada Cystic Fibrosis



Analysis 1.1. Comparison 1 Oral nonsteroidal anti-inflammatory drug versus placebo, Outcome 1 Annual rate of change in % predicted FEV_1.

Study or subgroup	NSAID N	NSAID Mean(SD)	Placebo N	Placebo Mean(SD)	Weight	Mean Difference Fixed, 95% CI
Konstan 1995	41	-2.2 (3.7)	43	-3.6 (3.6)	50.88%	1.43[-0.12,2.98]
Lands 2007	70	-1.5 (4.8)	72	-2.7 (4.8)	49.12%	1.2[-0.38,2.78]
Total ***	**111**		**115**		**100%**	**1.32[0.21,2.42]**

Heterogeneity: Tau²=0; Chi²=0.04, df=1(P=0.84); I²=0%
Test for overall effect: Z=2.33(P=0.02)

Analysis 1.2. Comparison 1 Oral nonsteroidal anti-inflammatory drug versus placebo, Outcome 2 Annual rate of change in % predicted FEV_1 (split by age).

Study or subgroup	NSAID N	NSAID Mean(SD)	Placebo N	Placebo Mean(SD)	Weight	Mean Difference Fixed, 95% CI
1.2.1 Under 13 years at randomisation						
Konstan 1995	24	-1.5 (3.8)	25	-4.2 (3.8)	26.88%	2.71[0.6,4.82]
Lands 2007	45	-2 (4.6)	53	-2.4 (4.7)	35.03%	0.42[-1.42,2.26]
Subtotal ***	**69**		**78**		**61.91%**	**1.41[0.03,2.8]**
Heterogeneity: Tau²=0; Chi²=2.57, df=1(P=0.11); I²=61.08%						
Test for overall effect: Z=2(P=0.05)						
1.2.2 13 years or over at randomisation						
Konstan 1995	17	-3.1 (3.2)	18	-2.8 (3.2)	26.18%	-0.36[-2.49,1.77]
Lands 2007	25	-0.4 (5.1)	19	-3.6 (5.5)	11.91%	3.2[0.04,6.36]
Subtotal ***	**42**		**37**		**38.09%**	**0.75[-1.02,2.52]**
Heterogeneity: Tau²=0; Chi²=3.34, df=1(P=0.07); I²=70.08%						
Test for overall effect: Z=0.83(P=0.4)						
Total ***	**111**		**115**		**100%**	**1.16[0.07,2.25]**

Heterogeneity: Tau²=0; Chi²=6.24, df=3(P=0.1); I²=51.95%
Test for overall effect: Z=2.09(P=0.04)
Test for subgroup differences: Chi²=0.33, df=1 (P=0.56), I²=0%

Analysis 1.3. Comparison 1 Oral nonsteroidal anti-inflammatory drug versus placebo, Outcome 3 Annual rate of change in % predicted FVC.

Study or subgroup	NSAID N	NSAID Mean(SD)	Placebo N	Placebo Mean(SD)	Weight	Mean Difference Fixed, 95% CI
Konstan 1995	41	-2 (3.3)	43	-3 (3.4)	50.19%	0.99[-0.43,2.41]
Lands 2007	70	-0.1 (4.3)	72	-1.6 (4.4)	49.81%	1.55[0.12,2.98]
Total ***	**111**		**115**		**100%**	**1.27[0.26,2.28]**

Heterogeneity: Tau²=0; Chi²=0.3, df=1(P=0.59); I²=0%
Test for overall effect: Z=2.47(P=0.01)

Favours Placebo -5 -2.5 0 2.5 5 Favours NSAIDs

Desain: Systematic review + meta-analysis

Populasi: 4 RCT, 287 pasien (usia 5–39 tahun), follow-up hingga 4 tahun
3 RCT membandingkan ibuprofen vs plasebo

Intervensi: NSAID oral (ibuprofen) minimal ≥2 bulan vs plasebo

Hasil:FEV1 %pred/ tahun: MD +1,32 (95% CI 0,21–2,42)
FVC %pred/ tahun: MD +1,27 (95% CI 0,26–2,28)
Efek lebih nyata pada anak (mis. FEV1 MD +1,41%/tahun, 95% CI 0,03–2,80)

Sullivan JE, Farrar HC. Fever and Antipyretic Use in Children. Pediatrics. 2011;127(3). doi:10.1542/peds.2010-3852

Paracetamol & Ibuprofen

TABLE 1 Antipyretic Information

Variable	Acetaminophen	Ibuprofen
Decline in temperature, °C	1–2	1–2
Time to onset, h	<1	<1
Time to peak effect, h	3–4	3–4
Duration of effect, h	4–6	6–8
Dose, mg/kg	10–15 every 4 h	10 every 6 h
Maximum daily dose, mg/kg	90 mg/kg[a]	40 mg/kg
Maximum daily adult dose, g/d	4	2.4
Lower age limit, mo[b]	3	6

Data represent approximate averages from referenced sources.[42,43,52,54,71,82]

[a] Label is for 75 mg/kg; 90 mg/kg per day should be limited to less than 3 consecutive days.[83,85]

[b] Unless specifically recommended by a health care provider for the younger patient and, then, only after the infant has been examined by a health care provider.

De la Cruz-Mena JE, Veroniki AA, Acosta-Reyes J, et al. Short-term Dual Therapy or Mono Therapy With Acetaminophen and Ibuprofen for Fever: A Network Meta-Analysis. Pediatrics. 2024;154(4):e2023065390. doi:10.1542/peds.2023-065390

Ibuprofen Alternating / Combined untuk Demam Anak

Outcome		Certainty*	Classification	Intervention	Relative effect estimates; OR (95% CI)	P-score	Absolute effect estimates. MD or RD (95% CI)	NNT (95% CI)
Primary	Fever Clearance	High	Not effective (Similar to acetaminophen)	Ibuprofen low dose	--	0.37	1.91 (-87.26; 83.42)	
		Low	May be not effective. (May be similar to acetaminophen)	Ibuprofen high dose	--	0.75	-32.82 (-68.68; 3.03)	
				Combined	--	0.62	-27.58 (-105.02; 49.86)	
		Comparator		Acetaminophen	--	0.25	--	
		Without evidence for this outcome		Alternating	--	--	--	
	Afebrile 4h	High	The most effective	Combined	**13.24 (4.59; 38.15)**	**0.99**	**-0.34 (-0.37; -0.26)**	**3 (4 ; 3)**
			Very effective	Alternating	**3.59 (1.77; 7.26)**	**0.67**	**-0.24 (-0.31; -0.12)**	**4 (8; 3)**
			Effective	Ibuprofen high dose	**1.76 (1.20; 2.57)**	**0.34**	**-0.12 (-0.19; 0.04)**	**8 (23; 5)**
		Comparator		Acetaminophen	--	0.00	--	
		Without evidence for this outcome		Ibuprofen low dose	--	--	--	
	Afebrile 6h	High	The most effective	Combined	**5.28 (2.38; 11.71)**	**0.84**	**-0.31 (-0.37; -0.19)**	**3 (5; 3)**
			Not effective (similar to acetaminophen)	Ibuprofen High dose	1.02 (0.66; 1.58)	0.18	0 (-0.11; 0.10)	--
		Low	May be among the most effective	Alternating	**5.06 (2.38; 11.71)**	**0.82**	**-0.30 (-0.39; 0.11)**	**3 (9; 3)**
		Comparator		Acetaminophen	--	0.16	--	
		Without evidence for this outcome		Ibuprofen low dose	--	--	--	
Secondary	Temp 4h	Low	May be the most effective	Combined	--	0.97	**-0.72 (-1.07; -0.38)**	
				Alternating	--	0.72	**-0.48 (-0.81; -0.14)**	
				Ibuprofen high dose	--	0.50	**-0.28 (-0.45; -0.12)**	
			Not effective (similar to acetaminophen)	Ibuprofen low dose		0.23	-0.10 (-0.43; 0.24)	
		Comparator		Acetaminophen	--	0.07	--	
	Temp 6h	High	The most effective	Alternating	--	**0.98**	**-1.51 (-2.10; -0.92)**	
				Combined	--	**0.76**	**-1.11 (-1.51; -0.70)**	
			Effective	Ibuprofen high dose	--	**0.48**	**-0.25 (-0.44; -0.06)**	
			Not effective (similar to acetaminophen)	Ibuprofen low dose	--	0.14	0.01 (-0.28; 0.31)	
		Comparator		Acetaminophen	--	0.13	--	
	Severe AE	High	Similar to acetaminophen (safe)	Alternating	0.88 (0.32; 2.17)	0.57	0 (-0.04; 0.02)	
		Low	May be similar to acetaminophen (may be safe)	Ibuprofen high dose	1.09 (0.11; 10.68)	0.53	0 (-0.22; 0.03)	
				Ibuprofen low dose	0.92 (0.36; 2.37)	0.45	0 (-0.22; 0.03)	
		Comparator		Acetaminophen	--	0.45	..	
		Without evidence for this outcome		Combined	--	--	--	
	Mild AE	Low	Similar to acetaminophen (safe)	Alternating	1.09 (0.38; 3.13)	0.63	0.01 (-0.23; 0.10)	
				Ibuprofen high dose	2.16 (0.96; 4.87	0.43	0.14 (-0.34; 0.01)	
			May be similar to acetaminophen (may be safe)	Combined	1.28 (0.53; 3.06)	0.52	0.04 (-0.22; 0.08)	
				Ibuprofen low dose	1.35 (0.91; 2.02	0.13	0.14 (-0.34; 0.01)	
		Comparator		Acetaminophen	--	0.79	--	

FIGURE 3

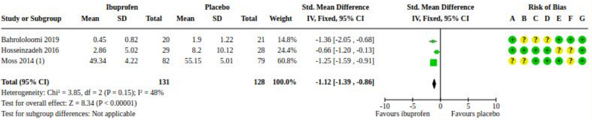
Results summary for the comparisons of interventions versus acetaminophen ranked using GRADE approach. Results are network meta-analysis effect estimates and 95% CIs for comparison against acetaminophen (common comparator). Numbers in **bold** are statistically significant results. *Overall GRADE certainty. Dark cell colors represent interventions with moderate- to high-quality evidence (high certainty on the results). Light cell colors represent very low- to low-quality evidence (low certainty on the results). Green tones indicate results different from comparator; orange tones indicate lack of differences between the intervention and comparator. ** Risk differences (RD) are calculated using baseline risk acquired from the placebo/control arm of included trials. CI, credible interval; MD, mean difference; OR, odds ratio. For dichotomous outcomes (afebrile fourth and sixth hours), an OR >1 indicates the treatment is superior to acetaminophen. For continuous outcomes, an MD <0 indicates the treatment is superior to acetaminophen. *** NNT are presented with 95% CI only for statistically significant results.

Desain: systematic review + network meta-analysis RCT pada anak demam

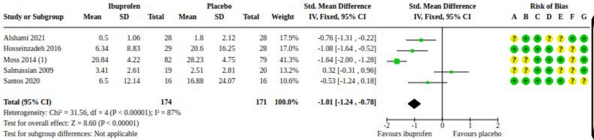
Populasi: 31 RCT (5009 anak)

Hasil:
Jam ke-4: combined dan alternating lebih unggul vs acetaminophen (OR 0.19 dan 0.20 vs acetaminophen)
TIdak ada perbedaan adverse events antara ibuprofen (dosis rendah/tinggi), alternating, combined vs acetaminophen

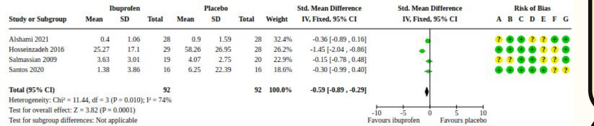
Analysis 1.1. Comparison 1: Ibuprofen versus placebo, Outcome 1: Pain intensity – as reported by the child, less than 2 hours postintervention



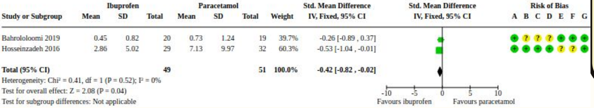
Analysis 1.2. Comparison 1: Ibuprofen versus placebo, Outcome 2: Pain intensity – as reported by the child, 2 hours to less than 24 hours postintervention



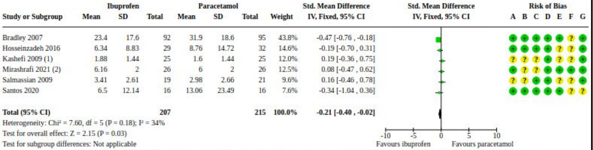
Analysis 1.3. Comparison 1: Ibuprofen versus placebo, Outcome 3: Pain intensity – as reported by the child, from 24 hours up to 7 days postintervention



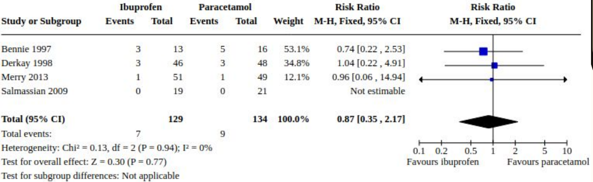
Analysis 2.1. Comparison 2: Ibuprofen versus paracetamol, Outcome 1: Pain intensity – as reported by the child, less than 2 hours postintervention



Analysis 2.2. Comparison 2: Ibuprofen versus paracetamol, Outcome 2: Pain intensity – as reported by the child, 2 hours to less than 24 hours postintervention



Analysis 2.8. Comparison 2: Ibuprofen versus paracetamol, Outcome 8: Number of children who required rescue medication less than 2 hours postintervention



Pessano S, Gloeck NR, Tancredi L, et al. Ibuprofen for acute postoperative pain in children. Cochrane Database Syst Rev. 2024;1(1):CD015432. Published 2024 Jan 5. doi:10.1002/14651858.CD015432.pub2

Ibuprofen untul Nyeri pada Anak

Desain: systematic review khusus anak dengan nyeri akut pascaoperasi

Populasi: 43 RCT 4265 anak, usia 2,2 bulan–17 tahun

Intervensi: Ibuprofen 10 mg/kg tiap 6 jam selama 24 jam, pre-op (tonsillectomy ± adenoidectomy, pre-op pada tindakan urologi (stent removal)

Hasil: ibuprofen vs plasebo menurunkan nyeri pada <2 jam (SMD -1,12) dan 24 jam (SMD -1,01)
Ibuprofen lebih baik vs paracetamol menurunkan intensitas nyeri pada <2 jam dan hingga 24 jam
Serious adverse events tidak dilaporkan, perbedaan perdarahan vs paracetamol kecil/tidak jelas

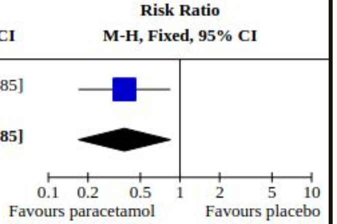
de Sévaux JLH, Damoiseaux RA, van de Pol AC, et al. Paracetamol (acetaminophen) or non-steroidal anti-inflammatory drugs, alone or combined, for pain relief in acute otitis media in children. Cochrane Database Syst Rev. 2023;8(8):CD011534. Published 2023 Aug 18. doi:10.1002/14651858.CD011534.pub3
Sullivan JE, Farrar HC. Fever and Antipyretic Use in Children. Pediatrics. 2011;127(3). doi:10.1542/peds.2010-3852

Paracetamol & Ibuprofen in Otitis Media Akut

Analysis 1.1. Comparison 1: Paracetamol versus placebo, Outcome 1: Pain at 48 hours

Study or Subgroup	Paracetamol Events	Paracetamol Total	Placebo Events	Placebo Total	Weight	Risk Ratio M-H, Fixed, 95% CI
Bertin 1996	7	73	19	75	100.0%	0.38 [0.17 , 0.85]
Total (95% CI)		**73**		**75**	**100.0%**	**0.38 [0.17 , 0.85]**
Total events:	7		19			

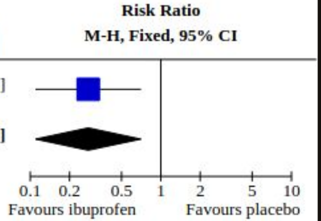
Heterogeneity: Not applicable
Test for overall effect: Z = 2.37 (P = 0.02)
Test for subgroup differences: Not applicable



Analysis 2.1. Comparison 2: NSAIDs versus placebo, Outcome 1: Pain at 48 hours

Study or Subgroup	Ibuprofen Events	Ibuprofen Total	Placebo Events	Placebo Total	Weight	Risk Ratio M-H, Fixed, 95% CI
Bertin 1996	5	71	19	75	100.0%	0.28 [0.11 , 0.70]
Total (95% CI)		**71**		**75**	**100.0%**	**0.28 [0.11 , 0.70]**
Total events:	5		19			

Heterogeneity: Not applicable
Test for overall effect: Z = 2.70 (P = 0.007)
Test for subgroup differences: Not applicable



Uji klinis menunjukkan parasetamol maupun ibuprofen (monoterapi) cenderung lebih efektif daripada plasebo untuk mengurangi nyeri telinga jangka pendek pada anak dengan OMA

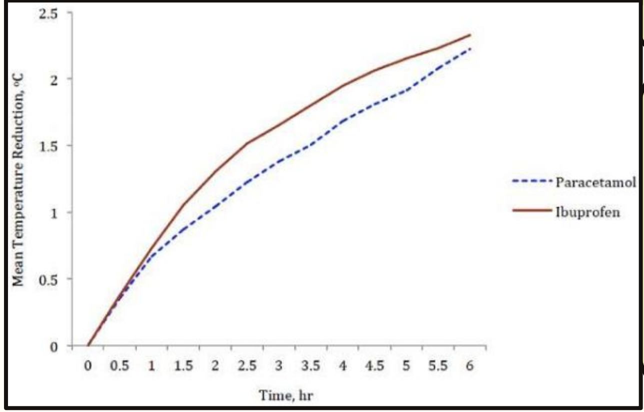
AAP tentang antipiretik mendukung penggunaan parasetamol atau ibuprofen untuk meningkatkan kenyamanan anak (comfort) dengan prinsip dosis berbasis berat badan

Ekaete Olajide Alaje et al. Ibuprofen versus paracetamol for treating fever in preschool children in Nigeria: a randomized clinical trial of effectiveness and safety. Pan African Medical Journal. 2020;36:350. [doi: 10.11604/pamj.2020.36.350.21393]

Ibuprofen vs Paracetamol pada Demam Anak

Table 2: comparison of mean temperature reduction at different time points

	Mean temperature reduction ± SD, °C				
Time, hr	Paracetamol group (n=70)	Ibuprofen group (n=70)	t- value	95% CI	p-value
0.5	0.36 ± 0.35	0.38 ± 0.38	-0.35	-0.14, 0.10	0.75
1	0.67 ± 0.47	0.73 ± 0.53	-0.74	-0.23, 0.10	0.48
1.5	0.87 ± 0.58	1.05 ± 0.61	-1.81	-0.39, 0.02	0.08
2	1.04 ± 0.66	1.30 ± 0.67	-2.31	-0.48, -0.04	0.02
2.5	1.23 ± 0.65	1.51 ± 0.67	-2.59	-0.51, -0.07	0.01
3	1.38±0.67	1.66 ± 0.72	-2.35	-0.51, -0.04	0.02
3.5	1.51 ± 0.64	1.80 ± 0.75	-2.52	-0.53, -0.06	0.02
4	1.68 ± 0.63	1.95 ± 0.76	-2.25	-0.50, -0.03	0.02
4.5	1.81 ± 0.65	2.06 ± 0.79	-2.06	-0.50, -0.01	0.04
5	1.91 ± 0.70	2.15 ± 0.82	-1.88	-0.50, 0.01	0.06
5.5	2.08 ± 0.75	2.23 ± 0.82	-1.12	-0.41, 0.11	0.26
6	2.22 ± 0.77	2.33 ± 0.87	-0.77	-0.38, 0.17	0.43



Desain: RCT terkontrol, terapi dosis tunggal ibuprofen vs paracetamol

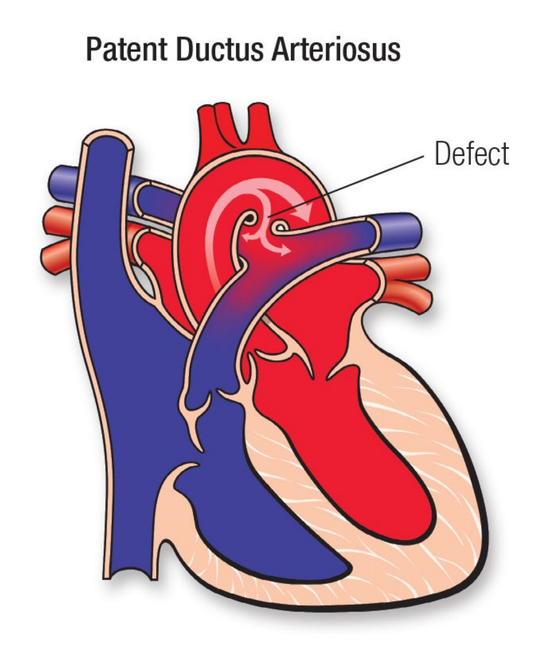
Populasi: 140 anak usia 6–59 bulan dengan suhu timpani 38–40°C

Intervensi: ibuprofen 10 mg/kg sekali vs paracetamol 15 mg/kg sekali (keduanya bentuk sirup)

Hasil: Penurunan demam pada ibuprofen lebih baik dibanding paracetamol
Proporsi afebrile pada 1.5–2.5 jam berbeda signifikan (p=0.04)
Efek samping ringan dan sebanding, muntah es paling sering

Pugnaloni F, Doni D, Lucente M, Fiocchi S, Capolupo I. Ductus Arteriosus in Fetal and Perinatal Life. Journal of Cardiovascular Development and Disease. 2024; 11(4):113. https://doi.org/10.3390/jcdd11040113
Jpnim.com. Published 2026. Accessed April 14, 2026. https://jpnim.com/plugins/generic/hypothesis/pdf.js/viewer/web/viewer.html?file=https%3A%2F%2Fjpnim.com%2Findex.php%2Fjpnim%2Farticle%2Fdownload%2F030226%2F170%2F640

Additional Effect of Ibuprofen in PDA



Patent Ductus Arteriosus (PDA) adalah shunt fetal antara arteri pulmonalis ↔ aorta yang normalnya menutup setelah lahir.

Setelah lahir: ↑ oksigen + perubahan sirkulasi → konstriksi otot polos DA (functional closure) → lalu remodeling (anatomic closure)

Prostaglandin (Termasuk PGE_2) membantu mempertahankan DA tetap terbuka pada masa fetal

PGE_2 bersifat vasodilator DA → mempertahankan patensi

Ibuprofen = inhibitor COX non-selektif → ↓ sintesis prostaglandin (termasuk PGE_2) → konstriksi DA → peluang penutupan ↑↑

Luo H, He J, Xu X, Chen H, Shi J. The impact of the route of administration on the efficacy and safety of the drug therapy for patent ductus arteriosus in premature infants: a systematic review and meta-analysis. PeerJ. 2024;12:e16591. Published 2024 Jan 29. doi:10.7717/peerj.16591

Ibuprofen Pada PDA

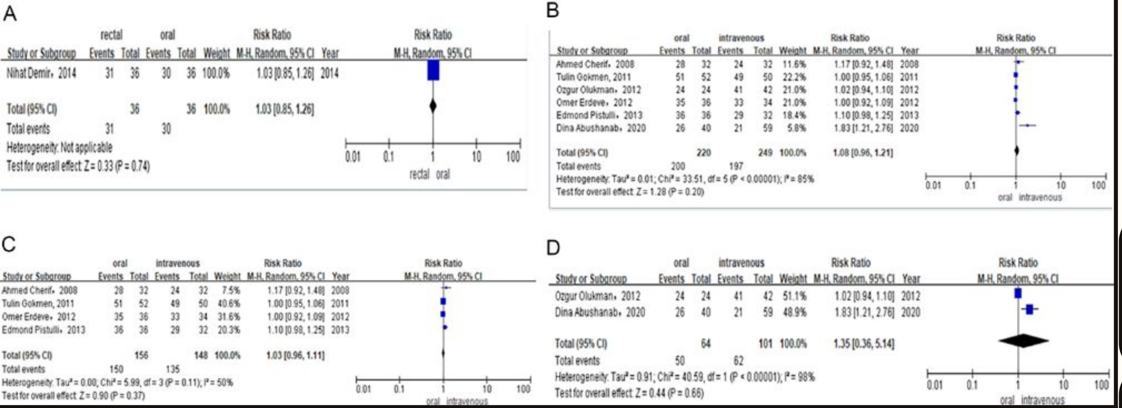


Figure 3 (A) Risk difference of RCTS of PDA closure rate after the first course of ibuprofen. (B) Risk difference of combined of RCTS and observational studies of PDA closure rate after total course of ibuprofen. (C) Risk difference of RCTS of PDA closure rate after total course of ibuprofen. (D) Risk difference of observational studies of PDA closure rate after total course of ibuprofen.

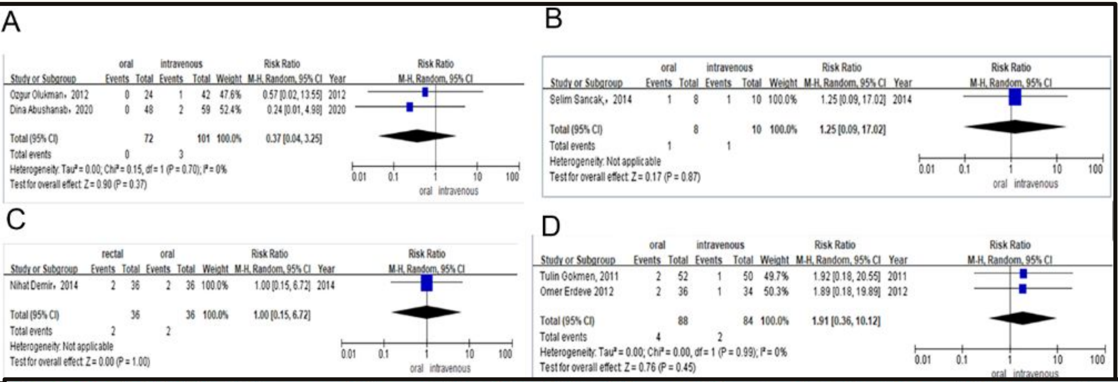


Figure 5 (A) Risk difference of observational studies of the need for surgical ligation of PDA after the ibuprofen treatment. (B) Risk difference of observational studies of the need for surgical ligation of PDA after the paracetamol treatment. (C) Risk difference of RCTs of the need for surgical ligation of PDA after the ibuprofen treatment. (D) Risk difference of RCTs of mortality during the ibuprofen therapy.

Desain: systematic review dan meta-analysis (6 RCT + 5 studi observasional)

Populasi: 630 neonatus prematur

Intervensi: oral ibuprofen vs iv ibuprofen untuk closure PDA

Hasil: Perbedaan signifikan tingkat penutupan PDA antara kelompok ibuprofen oral dibandingkan dengan kelompok ibuprofen intravena (RR = 1,27, 95%; CI [1,13–1,44]; P<0,0001)
Todak ada peningkatan signifkan di adverse events / kebutuhan tindakan bedah

Mitra S, Florez ID, Tamayo ME, et al. Association of Placebo, Indomethacin, Ibuprofen, and Acetaminophen With Closure of Hemodynamically Significant Patent Ductus Arteriosus in Preterm Infants. JAMA. 2018;319(12):1221. doi:https://doi.org/10.1001/jama.2018.1896

High-Dose Oral Ibuprofen Paling Kuat Untuk Closure

Desain: systematic review dan network meta-analysis dari 68 RCT

Populasi: 4.802 bayi prematur dengan PDA

Intervensi: placebo/no treatment, indomethacin, ibuprofen IV/oral, high-dose oral ibuprofen, acetaminophen

Hasil: High-dose oral ibuprofen closure lebih tinggi dibanding standard IV ibuprofen (OR 3.59) & IV indomethacin (OR 2.35)

A Patent ductus arteriosus closure, $P = .07$ for network inconsistency[a]

Ibuprofen, high oral dose Mean SUCRA, 0.89 (SD, 0.12); median rank, 2 (95% CrI, 1-5)									
0.98 (0.20-4.24)	**Ibuprofen, high IV dose** Mean SUCRA, 0.84 (SD, 0.20); median rank, 2 (95% CrI, 1-7)								
1.23 (0.62-2.48)	1.25 (0.31-5.77)	**Acetaminophen, oral** Mean SUCRA, 0.82 (SD, 0.12); median rank, 3 (95% CrI, 1-5)							
1.63 (0.84-3.24)	1.66 (0.45-7.07)	1.33 (0.81-2.17)	**Ibuprofen, standard oral dose** Mean SUCRA, 0.68 (SD, 0.10); median rank, 4 (95% CrI, 2-6)						
2.35 (1.08-5.31)	2.41 (0.68-9.86)	1.92 (1.00-3.68)	1.45 (0.94-2.24)	**Indomethacin, IV** Mean SUCRA, 0.48 (SD, 0.1); median rank, 6 (95% CrI, 4-7)					
2.36 (1.04-5.46)	2.42 (0.64-10.35)	1.93 (0.95-3.84)	1.46 (0.87-2.37)	1.01 (0.64-1.52)	**Indomethacin, other types** Mean SUCRA, 0.47 (SD, 0.13); median rank, 6 (95% CrI, 3-8)				
2.84 (0.85-9.59)	2.91 (0.62-15.44)	2.31 (0.76-7.05)	1.74 (0.64-4.77)	1.20 (0.47-3.10)	1.19 (0.44-3.40)	**Indomethacin, continuous IV infusion** Mean SUCRA, 0.40 (SD, 0.21); median rank, 7 (95% CrI, 2-9)			
3.59 (1.64-8.17)	3.68 (1.09-14.59)	2.93 (1.53-5.62)	2.22 (1.44-3.40)	1.53 (1.13-2.09)	1.51 (0.95-2.56)	1.27 (0.50-3.19)	**Ibuprofen, standard IV dose** Mean SUCRA, 0.24 (SD, 0.07); median rank, 8 (95% CrI, 7-9)		
5.01 (1.56-17.00)	5.19 (1.13-26.09)	4.08 (1.35-12.47)	3.08 (1.16-8.35)	2.14 (0.84-5.50)	2.12 (0.79-5.99)	1.79 (0.49-6.24)	1.39 (0.58-3.41)	**Ibuprofen, continuous IV infusion** Mean SUCRA, 0.17 (SD, 0.13); median rank, 9 (95% CrI, 5-9)	
16.12 (7.25-37.34)	16.53 (4.50-70.42)	13.16 (6.75-26.26)	9.93 (6.23-16.08)	6.88 (4.62-10.28)	6.82 (4.21-11.41)	5.71 (2.07-15.60)	4.49 (2.90-6.95)	3.23 (1.20-8.58)	**Placebo or no treatment** Mean SUCRA, 0.001 (SD, 0.012); median rank, 10 (95% CrI, 10-10)

The unlabeled data in the boxes are odds ratios (ORs) and 95% credible intervals (CrIs). An OR >1 suggests that the upper left treatment is associated with a higher odds of having the outcome of interest vs the corresponding lower right treatment and the opposite is true for an OR <1. SUCRA, surface under the cumulative ranking curve.

[a] $P < .05$ indicates statistically significant inconsistency between the direct and indirect estimates in the network as assessed by the design × treatment interaction model.

[b] No studies using a continuous IV infusion of ibuprofen reported on the need for repeat pharmacotherapy; therefore, this intervention was excluded from the respective outcome network.

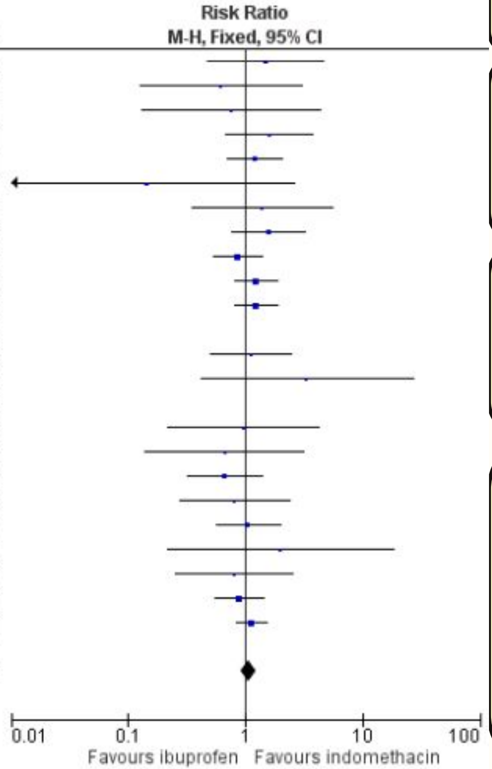
Ohlsson A, Walia R, Shah SS. Ibuprofen for the treatment of patent ductus arteriosus in preterm or low birth weight (or both) infants. Cochrane Database Syst Rev. 2020;2(2):CD003481. Published 2020 Feb 11. doi:10.1002/14651858.CD003481.pub8

Ibuprofen vs Indometacin pada PDA

Figure 6. Forest plot of comparison: 3 Intravenous or oral ibuprofen versus intravenous or oral indomethacin, outcome: 3.1 Failure to close a patent ductus arteriosus (after single or three doses).

Study or Subgroup	Ibuprofen Events	Ibuprofen Total	Indomethacin Events	Indomethacin Total	Weight	Risk Ratio M-H, Fixed, 95% CI
Adamska 2005	5	16	4	19	1.6%	1.48 [0.48, 4.61]
Akisu 2001	2	12	3	11	1.4%	0.61 [0.12, 3.00]
Aly 2007	2	12	2	9	1.0%	0.75 [0.13, 4.36]
Chotigeat 2003	8	15	5	15	2.2%	1.60 [0.68, 3.77]
El-Mashad 2017	23	100	19	100	8.5%	1.21 [0.70, 2.08]
Fakhraee 2007	0	18	3	18	1.6%	0.14 [0.01, 2.58]
Gimeno Navarro 2005	4	23	3	24	1.3%	1.39 [0.35, 5.55]
Hammerman 2008	13	32	8	31	3.6%	1.57 [0.76, 3.26]
Lago 2002	25	94	25	81	12.0%	0.86 [0.54, 1.38]
Lin 2012	30	71	25	73	11.0%	1.23 [0.81, 1.87]
Lin 2017	30	71	25	73	11.0%	1.23 [0.81, 1.87]
Mosca 1997	0	8	0	8		Not estimable
Patel 1995	8	18	6	15	2.9%	1.11 [0.50, 2.49]
Patel 2000	4	18	1	15	0.5%	3.33 [0.42, 26.72]
Pezzati 1999	0	9	0	8		Not estimable
Plavka 2001	3	21	3	20	1.4%	0.95 [0.22, 4.18]
Pourarian 2008	2	10	3	10	1.3%	0.67 [0.14, 3.17]
Salama 2008	7	21	10	20	4.6%	0.67 [0.32, 1.41]
Su 2003	5	32	6	31	2.7%	0.81 [0.27, 2.38]
Su 2008	15	60	14	59	6.3%	1.05 [0.56, 1.99]
Supapannachart 2002	2	9	1	9	0.4%	2.00 [0.22, 18.33]
Van Overmeire 1997	4	20	5	20	2.2%	0.80 [0.25, 2.55]
Van Overmeire 2000	22	74	25	74	11.1%	0.88 [0.55, 1.41]
Yadav 2014	34	48	22	35	11.3%	1.13 [0.82, 1.54]
Total (95% CI)		**812**		**778**	**100.0%**	**1.07 [0.92, 1.24]**
Total events	248		218			

Heterogeneity: Chi² = 11.42, df = 21 (P = 0.95); I² = 0%
Test for overall effect: Z = 0.92 (P = 0.36)



Desain: systematic review Cochrane

Populasi: 39 trial, total 2.843 bayi prematur/low birth weight

Intervensi: ibuprofen dibanding indomethacin atau placebo/no treatment

Hasil: Ibuprofen sama efektifnya dengan indomethacin untuk closure PDA
Tapio dibanding indomethacin, ibuprofen menurunkan risiko necrotizing enterocolitis dan transient renal insufficiency

Pokharkar V, Sajith M, Vallet T, et al. Acceptability of different oral dosage forms in paediatric patients in hospital setting. Archives of Disease in Childhood. Published online November 19, 2021:archdischild-2021-322604. doi:https://doi.org/10.1136/archdischild-2021-322604
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Ibuprofen Suspention vs Tablet

Anak kecil sering belum bisa menelan tablet

↓

tablet jadinya dibelah, dihancurin jadi serbuk atu bubuj, dicampur makanan/minuman, intinya diubah caranya sebelum diberikan ke anak

↓

Pemberian obat jadi ga praktis + berpotensi mengganggu akurasi dosis + ga konsisten pemberiannya

Suspensi ibuprofen memberi keuntungan praktis yang sangat relevan di pediatri:

- Dosis lebih gampang disesuaikan mg/kgBB & lebih presisi
- Lebih gampang diberikan pada anak yang ga suka tablet
- Tidak perlu manipulasi tablet
- Lebih cocok untuk anak usia muda yang acceptability-nya masih rendah terhadap tablet

Oral liquid lebih diterima pada anak kecil

- Studi acceptability pediatrik menyatakan oral liquids positively accepted pada infant dan toddler.
- Sedangkan acceptability tablet baru meningkat seiring usia mulai usia 6 tahun

Memanipulasi tablet sering terjadi

- Studi rumah sakit anak di Belanda, dari 571 obat, terdapat 169 manipulasi (30%) dan cuma 55% manipulasi yang sesuai instruksi resmi sisanya manupulasi tanpa instruksi
- Pada observasi perawat, 60% melakukan manipulasi, dan 52% di antaranya ga sesuai protokol

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Viktorio Mihaylov, Mihaela Tosheva, Petrov V, Stefka Titeva. Technological approaches to increase the bioavailability of Ibuprofen. Pharmacia. 2025;72:1-11. doi:https://doi.org/10.3897/pharmacia.72.e149517
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Ibuprofen Micronized (25 μm)

Micronization = proses mengecilkan ukuran partikel ibuprofen ke skala mikrometer (μm) Microparticles = 10^{-7}–10^{-4} m / 0.1–100 μm
25 μm = micro partikel (satuan μm) dibadingkan raw/coarse yang jauh lebih besar (180 μm, 144±87 μm, 71.3 μm)
Ibuprofen = BCS Class II (kelarutan rendah, permeabilitas tinggi) → sering dissolution-limited → harus larut dulu sebelum cepat diserap → dissolution rate memengaruhi absorption rate & onset
Ukuran partikel ↓ → luas permukaan ↑ → dissolution ↑ → lebih cepat tersedia untuk diserap → Tmax bisa lebih cepat
Formulasi yang llebih cepat available untuk diserap → capaii Tmax juga lebih cepat

Randomized Controlled Trial > Int J Clin Pharmacol Ther. 2015 Nov;53(11):972-9.
doi: 10.5414/CP202368.

Ibuprofen lysinate, quicker and less variable: relative bioavailability compared to ibuprofen base in a pediatric suspension dosage form

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PMID: 26249766 DOI: 10.5414/CP202368

Abstract

Objectives: To assess and compare the bioavailability of ibuprofen enantiomers (R and S) of two different pediatric suspensions: the first one with ibuprofen lysinate (Algidrin® Pediátrico, FARDI S.A., Barcelona, Spain) and the second one with ibuprofen base (Dalsy®, Abbott Laboratories S.A., Madrid, Spain).

Methods: A randomized, open-label, single-dose, balanced, crossover study under fasting conditions was performed at the CIM-Sant Pau. 24 healthy volunteers received a single dose of ibuprofen lysinate (AlgidrinÅ® PediÃ¡trico, FARDI S.A.) and ibuprofen base (Dalsy®, Abbott Laboratories S.A.) equivalent to 400 mg of ibuprofen. 18 blood samples were drawn and ibuprofen enantiomer plasma concentrations were determined using an enantioselective analytical method. An analysis of variance (ANOVA) model was used, and the 90% confidence intervals (CI) were calculated; further analyses were made regarding rate of absorption and variability.

Results: The pharmacokinetic parameters (Algidrin® Pediátrico vs. Dalsy® (Mean±SD)) were: S-enantiomer: Cmax=22.39±5.33 vs. 19.97±3.19 μg/mL; AUC0t=74.83±16.69 vs. 74.64±14.80 μg×h/mL, and AUC0∞=77.46±19.33 vs. 76.98±17.13 μg×h/mL; and for R-enantiomer: Cmax=21.74±3.76 vs. 15.20±2.03 μg/mL; AUC0t=57.55±10.17 vs. 46.13±9.61 μg×h/mL, and AUC0∞ value was 58.49±10.57 vs. 47.03±10.02 μg×h/mL. The tmax (Median) for S-enantiomer (active) were: 0.5 vs. 1.33 hours (p=0.001) and for R-enantiomer: 0.5 vs. 1.0 hours (p=0.004). Ibuprofen pharmacokinetic values may vary under fed state and in pediatric population.

Conclusions: While S-ibuprofen shows a similar bioavailability for AUC0t, AUC0∞, and Cmax, R-ibuprofen shows suprabioavailability for the lysinate formulation. The rate of absorption of the ibuprofen lysinate suspension is quicker and less variable than that of the ibuprofen base reference suspension and it exhibits a shorter tmax, which is of particular interest for achieving a rapid and homogeneous analgesic and antipyretic effect.

Ferrero-Cafiero JM, Gich I, Puntes M, et al. Ibuprofen lysinate, quicker and less variable: relative bioavailability compared to ibuprofen base in a pediatric suspension dosage form. Int J Clin Pharmacol Ther. 2015;53(11):972-979. doi:10.5414/CP202368

Formulasi Cepat Larut → Tmax Lebih Cepat

Desain: randomized, open-label, single-dose, crossover; n=24 (volunteer sehat), dosis ekuivalen 400 mg ibuprofen

Hasil:
median Tmax S-ibuprofen 0.5 jam vs 1.33 jam (p=0.001)
median Tmax R-ibuprofen 0.5 jam vs 1.0 jam (p=0.004)

Percepet larut untuk diserap → mempercepat Tmax (rate of absorption)

Scott CS, Retsch-Bogart GZ, Kustra RP, Graham KM, Glasscock BJ, Smith PC. The pharmacokinetics of ibuprofen suspension, chewable tablets, and tablets in children with cystic fibrosis. J Pediatr. 1999;134(1):58-63. doi:10.1016/s0022-3476(99)70387-2

Table II. Ibuprofen pharmacokinetic parameters*

	Suspension (n = 22)	Chewable tablet (n = 4)	Tablet (n = 12)
Dose (mg)	365 ± 111	337 ± 47	650 ± 193
Dose/kg (mg/kg)	20.3 ± 0.8	20.1 ± 1.3	21.7 ± 2.2
T_{max} (h)	0.74 ± 0.43§	1.50 ± 0.58	1.33 ± 0.95§
C_{max} (mg/L)	78.2 ± 23.9	55.6 ± 18.2	71.0 ± 24.0
C_{max} Norm† (mg/L)	76.9 ± 23.3	55.3 ± 16.9	65.2 ± 20.7
AUC Norm‡ (mg × min/mL)	9.2 ± 2.6	10.9 ± 2.3	12.1 ± 5.0
k_e (h^{-1})	0.56 ± 0.17	0.47 ± 0.15	0.49 ± 0.24
$T_{1/2}$ (h)	1.4 ± 0.5	1.6 ± 0.4	1.9 ± 1.4
V_d/F (L/kg)	0.27 ± 0.11	0.26 ± 0.10	0.26 ± 0.10
Cl/F (mL/min/kg)	2.33 ± 0.54	1.90 ± 0.44	1.88 ± 0.64

AUC, Area under the plasma concentration-time curve from time zero to infinity; k_e, apparent terminal disposition rate constant; $T_{1/2}$, terminal elimination half-life; V_d/F, apparent oral predicted volume of distribution during terminal phase.

*Values expressed as mean ± SD.

†C_{max} Norm = C_{max} normalized for dose = C_{max} ÷ Dose/kg × 20 mg/kg.

‡AUC Norm = AUC normalized for dose = AUC ÷ Dose/kg × 20 mg/kg.

§*P* < .02 (suspension vs tablet; ANOVA with post hoc Scheffe test).

PK Klinis Ibuprogfen Anak: Suspensi vs Tablet

Desain: clinical trial; 20 mg/kg; n=38 anak CF; bandingkan suspensi vs chewable vs tablet

Hasil:
Tmax beda signifikan (suspensi vs tablet)
Sampling puncak pada 30–60 menit karena puncak bisa terjadi cepat

Sediaan yang lebih cepat available di GI → puncak (Cmax) tercapai lebih cepat

When to Use Ibuprofen Syrup vs Paracetamol IV

Ibuprofen Syrup Preferred When:

- Anak mampu minum oral dan tidak muntah
- Demam/nyeri ringan–sedang dengan komponen inflamasi (OMA, tonsilofaringitis, myalgia)
- Dibutuhkan efek analgesik + antiinflamasi.
- Tidak ada kontraindikasi NSAID (dehidrasi berat, gangguan ginjal, risiko GI)

Paracetamol IV Preferred When:

- Anak tidak bisa minum oral (muntah, ileus, perioperatif)
- Nyeri atau demam pada kondisi rawat inap / pasca operasi
- Dibutuhkan onset cepat dan dosis presisi
- Risiko NSAID perlu dihindari

Take Home Messages

Demam dan nyeri anak→ gejala, terapi antipiretik-analgesik bertujuan mengurangi distress dan meningkatkan kenyamanan, bukan buat menurunkan angka suhu saja

Parasetamol dan ibuprofen sama-sama efektif dan aman bila digunakan sesuai dosis berbasis berat badan dan indikasi klinisnya

Ibuprofen sirup lebih unggul pada demam dan nyeri dengan komponen inflamasi bila anak mampu minum oral dan anaj tidak memiliki kontraindikasi NSAID

Ibuprofen dapat membantu meningkatkan tingkat penutupan PDA pada bayi

Paracetamol IV dipilih berdasarkan kondisi rute, terutama pada anak yang tidak bisa minum oral, pascaoperasi, atau memerlukan kontrol nyeri/demam di setting rawat inap

Terima Kasih